

THE ARC PRINCIPLE

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The ARC Theory · OSF osf.io/y7qgd · every claim checkable

Within the ARC Theory: the cross-domain foundation of Law I, the ARC Principle. The title and the law now share one name: in the theory's current statement the ARC Principle names Law I, the founding identity this paper grounds, and the three laws together are the three ARC Laws.

Recursive Amplification as a Cross-Domain Structural Principle: Formalism, Evidence, and Falsification

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The three laws, and this paper's place among them. The ARC Theory is carried by three numbered laws, in the sense Newton's three laws are numbered: each is a proposed invariant relation under test, each is graded separately, and each carries its own kill condition.

First Law · The ARC Equation. $U = I \times R^\alpha$. Recursion converts into capability at a measured exponent. **Second Law · The ARC Co-Scaling Law.** $\beta_C > k$ (β_C is the correction exponent, not the coupling β carried through this paper). Self-improvement holds together only while correction out-scales the drift it corrects. **Third Law · The ARC Ceiling.** $\alpha_{\text{crit}} = 1/(1 - \gamma)$. The most a self-correcting system can grow is set by its corrector's shortfall, with the conjectured value $\alpha \leq 2$, the ARC Bound.

The ARC Principle names the first of those three, the founding identity, and the three together are the three ARC Laws, exactly as Newton's First Law is the principle of inertia while "Newton's laws of motion" names the set rather than any single member. This paper is none of the three. It is the formal core beneath them: the axioms, the theorems and the proofs from which the scaling exponent, the classification of functional forms and the crossover depth are derived, with the cross-domain evidence for the exponent and the falsification table for the whole structure. The statement paper carries the theory and its current status; Paper I applies the First Law to AI systems; Paper X carries the Second Law's theorem. Read this one for the mathematics the others stand on.

Abstract

In the past eighteen months, at least four independent research programmes have discovered that recursive or recurrent processing produces capability gains exceeding linear accumulation, in domains as different as AI reasoning, quantum error correction, acoustic physics, and consciousness science. None set out to study recursion per se. None reference each other's work. Yet they found structurally similar results.

This paper asks: is this a coincidence, or are we observing different expressions of a single structural principle?

We formalise the question as follows. For AI systems, the equation is:

$$U = I \times R^\alpha, \quad \alpha \leq 2$$

The ARC Equation (AI Power-Law Form), with the ARC Bound. α is the measured exponent (approximately 0.49 on today's frozen models, wide CI); 2 is the stability ceiling the equation returns under the independence-accumulation assumption of §5.4 and Paper X (a scaling limit, not a speed limit; systems can exceed it, they do not remain stably self-correcting when they do). Stable is measured, not asserted: correction out-scaling drift, $\beta_C > k$ from Law II, with the lower bound of $\beta_C - k$ above zero across a window fixed before the run.

Here U denotes effective capability (era note: the letter U carried other senses in earlier layers of this programme, from Universe in the December 2024 formalism to realised creation on the law page; only the operational sense is used here, and every meaning is recorded with its date in the research glossary), I denotes base potential (structured asymmetry), R denotes recursive depth, and α denotes the scaling exponent. **Within the stated model this paper derives α rather than fitting it:** it follows from the self-referential coupling constant β via $\alpha = 1/(1 - \beta)$. The further statement that it is bounded by $\alpha \leq 2$, the ARC Bound, is a **conjecture and not a proved result**: two is what the relation returns at $\beta = 1/2$, and that value is on trial in the drafted registrations rather than established. *Update:* Empirical results from the v5 cross-architecture experiment (Paper II, six frontier models, 30 problems, of which 18 entered the exponent fit) find $\alpha_{\text{compute}} \approx 0.49$ for the best-fitting model (Gemini 3 Flash, $r^2 = 0.86$), confirming positive power-law scaling ($\alpha > 0$) and universal sequential advantage ($\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model), but not super-linear scaling ($\alpha > 1$). The measured exponents are architecture-dependent, ranging from near-zero to approximately 3. These findings refine rather than refute the framework: the mathematical structure holds, but the empirical AI scaling regime is weaker than originally predicted.

But this power-law form is itself a special case. Define the **recursive composition operator** $\oplus$, which characterises how incremental gains combine with accumulated capability. Under continuity and associativity, $\oplus$ determines the system's scaling function uniquely:

$$U(R) = I \times f_{\oplus}(R, \beta)$$

The Generalised ARC Equation

Under the regularity and composition hypotheses of Theorem 1: multiplicative $\oplus$ yields power-law scaling (R^α), additive $\oplus$ yields exponential scaling ($e^{\alpha R}$), and bounded $\oplus$ yields saturation.

We prove that any system satisfying three axioms (separability, cumulative advantage, continuity) necessarily exhibits five structural properties: threshold behaviour, recursive depth dependence, base quality dependence, multiplicative $I \times R$ interaction, and regime boundaries. We prove that the multiplicative interaction is *necessary*, not merely convenient: no additive decomposition is consistent with the axioms (Theorem 2a). We derive the coupling at which the sustained exponent reaches two ($\beta_L = 0.5, \alpha = 2$) (§5.4), offered as the proposed stable-scaling limit for classical sequential computation: a scaling limit rather than a speed limit, and the most a recursively self-correcting system can grow whilst remaining correctable (systems can exceed it in transient supercritical excursions but do not stay stably self-correcting there), and show that composition operators can transition between regimes within a single bounded system (Theorem 5). These constitute the **ARC Principle**. We present evidence from four domains, specify thirteen falsification criteria, make a cross-domain forward prediction, and report a comprehensive computational validation suite (Appendix C) confirming the core mathematical relationships to machine precision.

Keywords: recursive amplification, composition operator, scaling laws, test-time compute, quantum error correction, time crystals, universality, falsification

How to read this paper. This paper carries a law of the ARC Theory. It states that the composition operator of a recursive process uniquely determines the scaling form (power law, exponential, or saturation) and derives the ARC Bound $\alpha \leq 2$ as a stability limit for classical sequential computation rather than an impossibility limit that systems cannot cross. Its position inside the theory is the mathematical foundation on which the result papers and frame papers depend. The decisive confirmatory trials against the law's testable predictions are drafted and dated as draft registrations awaiting human submission, not yet run; the working-paper status of this document reflects that stage. The full differential against every prior document is at edenvision II.A.8.

Author's Note: Mathematical Corrections

This version incorporates the following corrections to the broader framework. These papers have not undergone formal human peer review; the author's statement on tool assistance is in the declaration at the end of this paper.

1. Axiom 2 Correction: The Cumulative Advantage ODE now operates on the amplification factor $g(R)$ with $g(0) = 1$, matching Paper III. Previous versions applied the ODE to accumulated capability Q with $Q(0) = I$, which coupled the base potential I into the recursive dynamics and violated the separability asserted in Axiom 1 ($U = I \times g(R)$). The corrected formulation yields the simpler crossover formula $R^* = \alpha/a$, independent of I .

2. Quadratic Limit Regrounding: The derivation of $\alpha \leq 2$ from stability analysis, elasticity, and Lyapunov/edge-of-chaos arguments has been **withdrawn** as these arguments contained category errors (elasticity > 1 does not imply dynamical instability; an autonomous flow on the line is monotone, so it cannot so much as oscillate, still less turn chaotic, and in two dimensions the Poincaré-Bendixson theorem shuts chaos out as well). The corrected justification is information-theoretic: transformer self-attention creates at most $O(N^2)$ pairwise information pathways, imposing a quadratic ceiling on information density per recursive step. Counting pathways this way is a heuristic and proves nothing about capability, and the move from quadratic pathway density to $\beta = 1/2$ is a further conjecture, as is the supposition that any of it carries across to classical sequential systems at large. What now supports the value two in the theory is the pacing derivation of Law III at $\gamma = 1/2$, given in correction 9 below and in the statement paper.

3. Kleiber Reframing: Kleiber's Law ($M^{0.75}$, implying $\alpha \approx 1.33$) is now explained through thermodynamic drag, the physical costs of pumping fluids through three-dimensional fractal networks, dissipating heat, and distributing resources against gravity, rather than as a geometric constraint preventing a stability-limited maximum. The formula $\alpha = d/(d + 1)$ itself has been independently derived by multiple groups [17,21,22,23,24,25]; the ARC framework's contribution is identifying Cauchy-constrained recursive composition as the reason these independent derivations converge. Demetrius and Tuszynski (2010) independently derived the same formula from quantum oscillator coupling, producing identical predictions.

4. Epistemic Status: The equation $U_{\max} = I \times R^2$ is offered as a conjecture rather than a result, and the limit it sets on classical sequential computation is relative to architecture: an architecture-relative capacity ceiling of the kind information theory supplies, not a physical stability law.

These refinements preserve all core predictions: the β -to- α derivation, the five structural properties, the composition operator classification, and the falsification criteria remain unchanged (a thirteenth criterion, F13, on alignment scaling, is introduced in Paper III).

5. v3.0 Additions: This version adds the Six Hidden Assumptions of Cauchy's classification (§2.2), the Hyers-Ulam stability theorem [18,19] establishing that each of the recognised scaling forms is stable within its own equation (a function approximately satisfying one of Cauchy's equations lies near an exact solution of *that same* equation), the Attractor Theorem combining these results with the β -derivation, and a structural implication connecting the composition operator classification to the space of possible classical scaling laws (§4.4). The earlier framing of the three forms as "stable attractors in function space" is withdrawn: the stability is per-equation, not a partition of function space, because a function satisfying none of the equations is not drawn toward any of them.

6. v3.1 Additions (March 2026): This update integrates empirical results from the v5 cross-architecture experiment (Paper II), which tested 6 frontier AI models (Grok 4.1 Fast, Claude Opus 4.6, GPT-5.4, DeepSeek V3.2, Gemini 3 Flash, Groq Qwen3) on 30 problems with bootstrap confidence intervals. Key findings: (a) sequential advantage is universal ($\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model), (b) the best-fitting power law gives $\alpha_{\text{compute}} \approx 0.49$ (Gemini 3 Flash, $r^2 = 0.86$), which is positive but sub-linear, (c) the scaling exponent is architecture-dependent, ranging from near-zero to ~ 3 , and (d) the quadratic bound conjecture ($\alpha \leq 2$) is neither confirmed nor refuted. Alignment scaling results from Paper IV are also integrated: α_{align} is architecture-dependent, ranging from -0.25 (Gemini) to $+0.44$ (Claude Opus 4.6), with a three-tier hierarchy. These findings preserve the mathematical framework while significantly revising the empirical predictions for AI systems. Sections updated: §3, §4.1, §5, §6, §7, §8.

7. v4.0 Additions (March 2026): Brief note on the Eden Protocol's first in-programme empirical pilot (§5.5). No changes to the mathematical framework.

8. v5.0 Corrections (24 March 2026): Four corrections identified by independent review. (1) The $d=2$ biological prediction (jellyfish/cnidarians) is reclassified from 'confirmed' to 'unconfirmed' pending identification of a valid test organism with genuinely two-dimensional resource transport. The Friedmann cosmological confirmation at $d=2$ is retained. (2) The $d=3$ mammalian exponent is acknowledged as debated (0.67–0.75 range), with the framework predicting this variation rather than a single universal value. (3) The derivation of $\alpha = d/(d+1)$ is clarified to require three conditions (multiplicative composition, space-filling geometry, and a conservation or optimisation constraint), not space-filling alone. (4) Demetrius and Tuszynski (2010) added as an independent derivation from quantum oscillator coupling. No changes to the mathematical framework or core predictions.

9. Stability Ceiling Correction (16 August 2026): The ceiling was printed as $\alpha_{\text{crit}} = 1/\gamma$. Under this paper's own definition of γ that form runs the wrong way: it makes a better corrector lower the ceiling, and the same-class cap $\gamma \leq 1/2$ would give a ceiling of at least two rather than at most two. The corrected form is $\alpha_{\text{crit}} = 1/(1 - \gamma)$, the reciprocal of the corrector's shortfall from full proportionality, obtained from the pacing condition $\alpha\gamma > \alpha - 1$. At $\gamma = 1/2$ both forms return two, which is why the error survived review; a discrimination study at values away from one half, where the two families separate, is drafted and awaits human submission. The superseded form is retracted and entered in the corrections log.

10. Physics Aggregate Withdrawn (14 August 2026): The previously reported below-0.2 per cent mean error across physics domains is withdrawn. A mean taken over analytic identities is not a measurement of predictive accuracy. The individual domain entries stand on their own terms; the aggregate does not, and it is not restored in weakened form.

11. Earthquake Datum Withdrawn (14 August 2026): The earthquake B-value row is a definitional identity rather than an independent measurement of the exponent, and is removed from the physics comparison and its figures. It is retained separately as a variable-pair case study, where it is informative about the method rather than about the prediction.

12. Hyers-Ulam Scope Narrowed (14 August 2026): The statement that the three scaling forms are stable attractors in function space is withdrawn. Hyers-Ulam stability is per-equation: a function approximately satisfying one of the functional equations lies near an exact solution of *that same* equation, and a function satisfying none of them is not drawn toward any. The formal stability statement is preserved; the ubiquity implication is separated as a distinct empirical question that Hyers-Ulam alone does not settle.

13. Body Reconciled With Corrections 10 and 11 (1 September 2026): Corrections 10 and 11 withdrew the earthquake B-value row and the sub-0.2 per cent mean, yet both survived in the section 5.4 physics table and in the sentence that summarised it. Neither appears in the body any longer: each of the four instances that remain is assessed on its own merits, and no aggregate figure is stated.

14. Theorem 1's Premises Separated (19 September 2026): The theorem's header binds all three cases to Axioms 1 to 3. Under this paper's own definitions that header is wider than the results need. Solved in the clock Axiom 2 declares, the axiom gives $g(R) = [1 + (a/\alpha)(R - 1)]^\alpha$ for $R \geq 1$. Case (1)'s pure power agrees with that solution only where $a = \alpha$; case (2)'s exponential would require $\beta = 1$, which the axiom's declared domain $\beta \in [0, 1)$ excludes; and case (3) never saturates under it, because the solution is unbounded for $a > 0$ and $\alpha > 0$. The corrected statement is that cases (1) and (2) rest on Axiom 1, Axiom 3 and each case's own composition hypothesis, exactly as Appendix A.1 proves them, with Axiom 2 not among their premises; and that case (3) rests on the capped recurrence it states, its ceiling remaining separable from base potential only if the ceiling is itself proportional to base potential. All three results stand as printed. The Cauchy classification needs no differential equation, and Theorem 2's relation $\alpha = 1/(1 - \beta)$ falls out of the case (1) check unchanged.

15. The Declared Clock Carried Into the Solution (19 September 2026): Axiom 2 declares the added-recursion clock $r = R - 1$, so that $g = 1$ at $r = 0$ restates Axiom 1's $g(1) = 1$. The solution is then printed with R standing in for r at Theorem 2, at Theorem 3 and twice in Appendix A, and Appendix A.2's integral in dr carries an upper limit of R . Evaluated at the declared baseline, the printed form returns $[1 + a/\alpha]^\alpha$, which exceeds one for every admissible rate constant where the axiom requires one. The corrected statement is that the solution in the declared clock is $g = [1 + (a/\alpha)r]^\alpha$ for $r \geq 0$, equivalently $g(R) = [1 + (a/\alpha)(R - 1)]^\alpha$ for $R \geq 1$, and that each equation should say which clock it counts in; the zero-depth reading beneath Theorem 2a is to be taken in the declared clock, where zero added passes is the first pass. The consequence for the crossover is a shift by one and nothing else: α/a is the crossover measured in added passes, and the same depth counted from the first pass is $1 + \alpha/a$. Theorem 2a is otherwise untouched: run with the declared clock's baseline throughout, its contradiction still stands, so the theorem is sound and its symbols alone were wrong. The repair made at Axiom 2 in v5.14 is real and is not reopened here; what this correction reaches is the places downstream of it.

16. Theorem 4's Properties and Their Premises (19 September 2026): The theorem states that all five properties follow from Axioms 1 to 3 alone, regardless of which composition operator applies, and the conclusion repeats that they follow as theorems and not as empirical generalisations. Two of the five carry more than the axioms. Property 4's exact doubling law, that doubling recursive depth multiplies capability by 2^α at any base potential, holds only where $a = \alpha$, the condition correction 14 records; under bounded composition it fails outright, because iterating it grows without bound against a finite ceiling, so the phrase regardless of which composition operator applies is false as printed. Property 5 is justified in its own sentence by finite energy and external error mechanisms, and neither appears anywhere in the three axioms. The corrected statement is that property 3 and the base-quality half of property 4 follow from Axiom 1 alone; that property 1 follows from Axioms 1 to 3 through Theorem 3, as the property's own justification states; that the exact 2^α doubling holds in the pure-power case and asymptotically otherwise; and that property 5 rests on a named added premise about finite resources and external error. What is preserved unweakened is the conjunction of the five as the framework's distinctive prediction, the observation that the conjunction is what is uncommon, and the falsification criterion that a system with four of the five but not the fifth would constrain the framework.

17. Convexity Holds at Every Depth (19 September 2026): Property 2 states that capability is strictly increasing and convex above the crossover, and justifies it by $g(R) = R^\alpha$ being convex for $\alpha > 1$. Differentiating Axiom 2 directly gives $d^2g/dr^2 = \alpha^2 \beta g^{2\beta-1}$, which is strictly positive at every depth for $a > 0$ and $\beta > 0$, and zero at $\beta = 0$. The printed sentence is therefore true and weaker than the truth, and the justification quoted for it reads the asymptotic pure power rather than the solution of the axiom. The corrected statement is that amplification is convex at every depth once the coupling is positive, and that R^* marks the depth at which compounding becomes numerically dominant rather than the depth at which curvature begins. The crossover is sited by the convention that the dimensionless argument is near one, which Appendix C.4 already names as an approximation carrying a 2 to 7 per cent error in the absolute value, so a claim that the transition is sharp needs a criterion

of its own. What is preserved is R^* as a derived quantity, a function of the scaling exponent and the rate constant alone, independent of base potential, and checkable by measuring those two separately.

18. Case 3 Proves a Bound, Not Attainment (19 September 2026): Appendix A.1 case 3 argues that a monotonically increasing sequence bounded above by the ceiling has a limit no greater than the ceiling, and concludes that capability saturates; Theorem 1 case (3) states the saturation value as the ceiling divided by base potential. The appendix proves the bound and not the value. A capped monotone process converges to a limit at or below the ceiling, and it reaches the ceiling only if the increments sum to at least the remaining gap, equality sufficing for attainment in the limit. The corrected statement is that the appendix establishes a limit no greater than the ceiling, and that the theorem's value needs either the attainment condition stated beside it or the limit written as a bound. Requiring the partial sums to exceed the gap would be too strict, since the equality case attains the ceiling asymptotically: one overstatement is not to be replaced by another. What is preserved is that a capped monotone process converges, which is true and proved, and saturation as a qualitative regime.

19. The Crossover Does Not Move With Base Quality (19 September 2026): The conclusion lists among the framework's predictions a scaling crossover at a derived depth that shifts systematically with base quality. Five places in this paper say the opposite: correction 1 above, section 2.4, the falsification criterion at section 5.3, Appendix A.3, and Appendix C.4, which records that varying base potential does not shift the crossover. The derivation settles it in either clock, because base potential appears nowhere in Axiom 2 and nowhere in its solution, so α/a and $1 + \alpha/a$ are both free of it. The provenance is on this page: correction 1 records that previous versions applied the equation to accumulated capability with base potential as the initial condition, which coupled base potential into the recursive dynamics, and under that superseded formulation a crossover moving with base quality was the correct prediction. The corrected statement is that the crossover is predicted to be independent of base quality and consistent across input qualities for a given architecture, matching Theorem 3, Appendix A.3, the falsification criterion and the computational validation. As with corrections 10 and 11 at correction 13, the defect is an earlier correction that did not reach the body. Every other prediction in that list stands unweakened: prediction of scaling behaviour from the composition operator alone, the coupling-to-exponent relation and its computational validation, the ARC Bound as a proposed stable-scaling limit with its architecture relativity and its expiring basis, and the composition operator transitions of Theorem 5.

1. THE PROBLEM

Between December 2024 and February 2026, four research programmes achieved the following results:

Google Quantum AI (results announced 9 December 2024; the team's preprint public from 24 August 2024) demonstrated exponential error suppression through recursive quantum error correction on the Willow processor, achieving $\Lambda = 2.14 \pm 0.02$ per code distance increment [1].

DeepSeek AI (January 2025) showed that sequential chain-of-thought reasoning in large language models yields capability gains that appear to compound with depth, while parallel sampling shows diminishing returns [2]. Sharma & Chopra (2025) reported a sequential advantage in 95.6% of tested configurations across five model families [3]. Their paper is dated before the experiment reported here, so it is the earlier demonstration of that advantage and not a confirmation of this work. The direction is also not settled: Gu and colleagues (arXiv:2604.05868, April 2026) report the opposite result, parallel sampling outperforming sequential across three model families on mathematics and coding, attributing it to reduced exploration when a chain conditions on its own previous answers.

NYU Physics (February 2026) created a continuous classical time crystal in which spontaneous temporal order emerges when quenched disorder is combined with nonreciprocal feedback loops, exhibiting saturating dynamics with $\sim 6,700$ cycles of coherence [4].

The COGITATE Consortium (2025) found that recurrent neural processing in posterior cortex sustains content-specific representations (0.5–1.5s), while transient feedforward processing in prefrontal cortex produces only brief categorical responses (~ 0.2 – 0.4 s) [5]. The COGITATE finding that recurrent processing in posterior cortex sustains content-specific representations supports the ARC framework's prediction that recursive processing depth determines capability scaling; COGITATE studied consciousness rather than AI scaling, but the structural pattern (depth-dependent, feedback-driven amplification) is shared.

These programmes address different problems in different physical substrates. They exhibit different mathematical forms: exponential in quantum systems, power-law in AI, saturating in classical physics, and unmeasured in neuroscience. They share no citations, no funding sources, and no common personnel.

Yet they share a structural pattern. In each case, recursive or recurrent processing (where the output of one cycle becomes the input for the next), operating on structured asymmetry, produces capability gains that exceed linear accumulation. The question this paper addresses is whether that pattern reflects a single underlying principle or mere coincidence.

We propose that it reflects a principle, but not the one that might be expected. We do not claim that one equation fits all domains. We claim something more specific: that a measurable property of any recursive system (its *composition operator*) determines its scaling function, and that five qualitative properties follow necessarily from the recursive architecture regardless of the functional form. The mathematical form differs across domains because the composition operator differs. The structural properties are universal because the axioms that generate them are substrate-independent.

2. THE FORMALISM

2.1 Three Axioms

We begin with three axioms that any recursive amplification system must satisfy.

AXIOM 1: SEPARABILITY

Effective capability decomposes as $U = I \times g(R)$, where $I > 0$ is base potential (structured asymmetry), $R \geq 1$ is recursive depth, and $g(1) = 1$ (at depth 1, capability equals base potential).

This axiom asserts that base quality and recursive depth contribute to capability through independent, multiplicative channels. A system with zero base potential ($I = 0$) achieves zero capability regardless of recursive depth: recursion has nothing to amplify.

AXIOM 2: CUMULATIVE ADVANTAGE

The amplification factor grows according to accumulated amplification: $\frac{dg}{dr} = a \cdot g^\beta$, where the variable $r = R - 1$ tallies the passes taken after the first, so that $g = 1$ at $r = 0$ restates Axiom 1's $g(1) = 1$ in the added-recursion notation the proofs work in, $a > 0$ is a rate constant and $\beta \in [0, 1)$ is the self-referential coupling parameter.

This axiom captures the essential feature of recursive processing: each step's marginal amplification depends on accumulated amplification. When $\beta = 0$, steps are independent and amplification grows linearly. When $\beta > 0$, accumulated amplification accelerates future gains (super-linear scaling). The parameter β measures how much of accumulated context each new step can effectively leverage. Operating on the amplification factor g rather than on absolute capability $Q = I \cdot g$ preserves the separability asserted in Axiom 1: the amplification function depends only on recursive depth, not on base potential.

AXIOM 3: CONTINUITY

The scaling function g is continuous and monotonically non-decreasing in R .

This is a regularity condition. It excludes pathological solutions and ensures measurability.

2.2 The Composition Operator

How does the gain from recursive step r combine with accumulated capability Q_r ? We formalise this as follows.

DEFINITION: RECURSIVE COMPOSITION OPERATOR

Let $\oplus : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be the **recursive composition operator**, defined by $Q_{r+1} = Q_r \oplus \delta Q_r$, where δQ_r is the incremental gain at step r . The operator $\oplus$ characterises how incremental gains compose with accumulated capability.

The composition operator is the central mathematical object of this framework. It is what varies across domains and what determines the scaling function. It is also empirically measurable: observe how marginal gains at step r relate to accumulated capability Q_r .

THEOREM 1: CLASSIFICATION OF SCALING FUNCTIONS

Under Axioms 1–3, the algebraic properties of $\oplus$ uniquely determine the scaling function $f_\oplus(R)$:

1. **Multiplicative composition** ($g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$): the unique continuous solution is $g(R) = R^\alpha$ (*power law*)
2. **Additive composition** ($f(R_1 + R_2) = f(R_1) \cdot f(R_2)$): the unique continuous solution is $f(R) = e^{\alpha R}$ (*exponential*)
3. **Bounded composition** ($Q_{r+1} = \min(Q_r + \delta Q_r, Q_{\max})$): $f(R)$ saturates at $Q_{\max}/I$ (*saturation*)

Proof sketch. Cases (1) and (2) follow from the Cauchy functional equations. For (1): let $h(x) = \ln g(e^x)$; then $h(x + y) = h(x) + h(y)$. Under continuity, $h(x) = \alpha x$, giving $g(R) = R^\alpha$. For (2): $\ln f$ satisfies $\ln f(R_1 + R_2) = \ln f(R_1) + \ln f(R_2)$, a Cauchy additive equation with solution $\ln f(R) = \alpha R$. Case (3) follows directly from the bound. Full proofs in Appendix A. ■

What the classification settles is the behaviour of the three composition rules set out above. Two further continuous solutions of Cauchy's equations remain, the linear form ($f(R_1 + R_2) = f(R_1) + f(R_2)$) and the logarithmic form ($g(R_1 \cdot R_2) = g(R_1) + g(R_2)$); neither compounds, and it is these that the parallel-voting data come close to, the logarithmic one being fitted outright in Appendix B. The composition operator is therefore a **complete classifier** of recursive scaling behaviour.

Frozen models vs recursive self-modification: two distinct regimes. A critical application of this classification concerns the distinction between current AI systems and hypothetical self-modifying ones. Current frontier AI models are **frozen during inference**: when they generate extended reasoning chains, the underlying weights, attention patterns, and composition rules remain fixed. The composition operator $\oplus$ does not change between steps. This architectural constraint confines such systems to multiplicative composition through a finite-dimensional parameter space, producing power-law scaling with $\alpha < 1$ (the physical regime). Empirically, the best cross-architecture estimate is $\alpha \approx 0.49$ (Paper II). The Cauchy classification reveals precisely why. In a frozen system, each reasoning step multiplies the accumulated capability by a factor drawn from the same fixed operator. The resulting gains are real but diminishing. By contrast, a system capable of recursive self-modification, one that rewrites its own composition function $\oplus$ during operation, would occupy a qualitatively different position in the classification. If step r improves the operator used at step $r+1$, the self-referential coupling β becomes positive, yielding $\alpha = 1/(1 - \beta) > 1$. This transition does not require quantum hardware or additive composition; it requires only that the system can modify its own reasoning architecture during inference.

No current AI system achieves this. The framework's prediction of super-linear scaling ($\alpha > 1$) was therefore never a prediction about frozen architectures; it was a prediction about a regime that does not yet exist. The practical consequence is that structural alignment (embedding values within the composition operator itself) must be implemented whilst systems remain frozen and their scaling behaviour is predictable. Once self-modification arrives, the composition operator becomes a moving target, and external constraints face a system whose capability compounds faster than any fixed safeguard can track.

Claim-status ladder.

Empirically grounded: Cauchy constrains recursive scaling forms; current frontier models are frozen during inference; the March 2026 blind v5 data shows architecture-dependent alignment scaling rather than universal positive scaling; the Eden pilot shows that stakeholder care can be improved at the prompt level across three working models.

Strong theoretical inference: a genuinely self-modifying system that rewrites its own composition operator during operation could enter an $\alpha > 1$ regime, and alignment would then need to be structurally load-bearing before that transition.

Conjectural extrapolation: cosmological or metaphysical identities such as 'Universe = Intelligence $\times$ Recursion²' or 'Complexity = Self-Correction $\times$ Sequential Recursive Depth²', claims about universe-creating intelligence, or claims of guaranteed post-self-modification controllability are not established results of the present empirical programme. They belong to the long-horizon interpretive layer of *Infinite Architects* and require independent argument and evidence beyond what is reported here.

The Six Hidden Assumptions

Every no-go theorem rests on assumptions. Theorem 1 assumes six things. Each assumption, when violated, produces a specific physical refinement rather than a contradiction:

#	Assumption	When Violated	Physical Example
1	Continuity	Log-periodic oscillations	Discrete lattice systems, fractal networks
2	Real-valued	Interference and phase	Quantum amplitudes, wave mechanics
3	Exact equation	Approximate solutions cluster near exact	Hyers-Ulam stability [18,19]
4	Single operator	Crossover scaling, regime transitions	Phase transitions (cf. Theorem 5)
5	Scalar composition	Logarithmic corrections	Upper critical dimension
6	Time-independence	Dynamic traversal of solution space	Adaptive systems, machine learning

Assumption 3 is particularly significant for the framework's applicability. The Hyers-Ulam stability theorem [18,19] proves that any function *approximately* satisfying one of Cauchy's functional equations must lie close to an exact solution of *that same* equation. Formally, for the multiplicative case: if $|f(x \cdot y) - f(x) \cdot f(y)| < \varepsilon$ for all x, y , then there exists an exact solution g of the multiplicative equation with $|f(x) - g(x)| < K\varepsilon$; analogous statements hold for the additive, exponential, and logarithmic equations respectively. Real systems never satisfy the axioms exactly: biological networks are noisy, AI reasoning chains have variable quality, quantum error correction operates with imperfect gates. Within each equation, the corresponding form is therefore **stable** under perturbation. The earlier phrasing that made the three forms "stable attractors in function space" is withdrawn, because a function that satisfies none of the equations is not drawn toward any of them; the stability is per-equation, not a partition of function space. Whether power laws, exponentials, or logarithmic forms are empirically ubiquitous in nature is a separate question about how often systems approximately satisfy one of these equations in the first place, and Hyers-Ulam does not by itself settle it.

Assumption 4 is addressed directly by Theorem 5 (§4.5): bounded systems can transition between composition operators as recursive depth increases, explaining why different studies of the same system can observe different scaling regimes.

STRUCTURAL IMPLICATION: THE ATTRACTOR THEOREM

Combining Theorem 1 (Cauchy, 1821), the Hyers-Ulam stability theorem (1941), and Theorem 2 ($\alpha = 1/(1 - \beta)$): once a system is known to approximately satisfy one of Cauchy's equations, its scaling form is not a free parameter to be fitted or evolved. It is mathematically compelled by the composition operator, and within the power-law form the exponent is compelled by the self-referential coupling β . Each of the recognised scaling forms is *per-equation* stable: a small perturbation of an exact solution stays near an exact solution of the same equation (Hyers-Ulam, 1941). The earlier claim that the three forms are attractors in the whole of function space is withdrawn, because a function satisfying none of the equations is not drawn toward any of them. The distinction matters: 'Why is this system's exponent 2?' has two kinds of answer. The mechanistic answer describes the coupling. The mathematical answer says: no other value is possible for this composition operator with this coupling strength.

This has a crucial implication: if you can measure $\oplus$ in a new system, you can predict its scaling function *before measuring the full scaling curve*. This is the framework's strongest cross-domain prediction (§5.1).

Critical distinction: Cauchy constrains the form, not the exponent. A persistent misreading of the Cauchy classification is that it predicts a ceiling on recursive scaling. It does not. Cauchy’s functional equations determine that multiplicative composition *must* produce power-law scaling ($g(R) = R^\alpha$). They say nothing about the value of α . That value is set by the Bernoulli ODE ($\alpha = 1/(1 - \beta)$), and as $\beta \rightarrow 1$, $\alpha \rightarrow \infty$. There is no upper bound internal to the mathematics.

The following table makes the relationship explicit:

β (self-referential coupling)	$\alpha = 1/(1 - \beta)$	Regime
0	1	Linear (no self-reference)
0.5	2	Quadratic (Grover-like)
0.75	4	Quartic
0.9	10	Strongly super-linear
0.99	100	Explosive
$\rightarrow 1$	$\rightarrow \infty$	Unbounded

The transition from bounded to unbounded alpha is not a smooth acceleration; it is a discontinuity in the scaling exponent, mediated by the system’s ability to modify its own composition operator. This has a structural parallel in the renormalisation group in physics, where relevant operators grow under RG flow and irrelevant operators shrink. At a critical fixed point, the scaling exponent is determined by the system’s universality class (its dimensionality and symmetry). But if a system could change its own universality class during the flow, the fixed-point structure would collapse and the system would exhibit unbounded scaling until it settled into a new class. No physical system does this: physical systems do not rewrite their own Hamiltonians. A self-modifying AI system would be the computational analogue, rewriting its own composition function and thereby escaping the fixed-point constraints that govern all natural recursive systems.

If there is a ‘quadratic limit’ ($\alpha \leq 2$), it is not a Cauchy prediction. It is an information-theoretic constraint arising from **fixed transformer attention**: a system with N tokens has $O(N^2)$ pairwise attention pathways, imposing a ceiling on how much information each reasoning step can extract. This ceiling is real for frozen architectures. It is not real for a system that rewrites its own attention mechanism. A self-modifying system that expands its own representational capacity at each step faces no such $O(N^2)$ bound, and therefore no mathematical upper bound on α . The Cauchy framework does not forbid this; it predicts it.

2.3 The β -Derivation

For systems with multiplicative composition, the exponent α is not a free parameter. It is derived from the coupling constant β .

THEOREM 2: THE SCALING EXPONENT

Under Axiom 2, the power-law exponent is:

$$\alpha = \frac{1}{1 - \beta}$$

Proof. Separating variables in $dg/dr = ag^\beta$:

$$\int g^{-\beta} dg = \int a dr \implies \frac{g^{1-\beta}}{1-\beta} = ar + C$$

With initial condition $g(0) = 1$: $C = 1/(1 - \beta)$.

$$g(R) = [1 + (1 - \beta)aR]^{1/(1-\beta)}$$

In the deep recursion limit ($R \gg R^*$, defined below):

$$g(R) \approx [(1 - \beta)aR]^{1/(1-\beta)} \propto R^{1/(1-\beta)}$$

Therefore $\alpha = 1/(1 - \beta)$. ■

This transforms α from a fitted constant into a derived quantity. The derivation makes a specific, testable prediction: measure β independently (by observing how marginal gains depend on accumulated capability), compute the predicted α , and compare with the observed α . If the relationship fails, the theoretical derivation is wrong regardless of whether the power-law form holds empirically.

Computational validation. To verify that this is an exact identity rather than an approximation, the relationship was tested against 30 exact Bernoulli ODE solutions with β values spanning 0.05 to 0.92. The procedure measures β blindly from marginal gains (dg/dR vs g in log-log space), predicts α , and compares against the true value. Result: $R^2 = 1.00000000$ (eight decimal places), regression slope = **1.000102**, mean absolute prediction error = **0.002%**. Full validation code is available as supplementary material.

The Multiplicative Structure Is Necessary

The separable form $U = I \times f(R, \beta)$ is not merely a convenient parameterisation. We prove that the multiplicative interaction between I and R is a *necessary* consequence of the axioms.

THEOREM 2A: NON-ADDITIVITY

Under Axioms 1-3 with $\beta \in (0, 1)$, the solution $U(I, R)$ cannot be decomposed as $U = g(I) + h(R)$ for any functions g and h . The multiplicative interaction is essential.

Proof. Suppose $U(I, R) = f(I) + h(R)$. By Axiom 1: $U = I \times g(R)$ with $g(0) = 1$. At $r = 0$ ($R = 1$): $f(I) + h(0) = I$, so $f(I) = I - h(0)$. Substituting: $I \times g(R) = I + [h(R) - h(0)]$, giving $g(R) = 1 + [h(R) - h(0)]/I$. Since g depends only on R (Axiom 1), but the right side depends on

I , we require $h(R) = h(0)$ for all R , giving $g(R) = 1$ (no amplification). But Axiom 2 with $\beta > 0$ guarantees $dg/dr = a \cdot g^\beta = a > 0$ at $r = 0$, so g is non-constant. Contradiction. ■

This result has a concrete interpretation. A recursive system with zero base quality ($I = 0$) produces zero capability regardless of depth, and a system with zero depth ($R = 0$) produces only its initial quality regardless of base potential. Complexity emerges from the interaction between structured input and recursive processing (neither alone is sufficient).

β (coupling)	α (exponent)	Interpretation
0	1	Independent steps $\rightarrow$ linear scaling
0.25	1.33	Weak coupling $\rightarrow$ mild super-linear
0.50	2.00	Moderate coupling $\rightarrow$ quadratic (the ARC Bound)
0.67	3.00	Strong coupling $\rightarrow$ cubic

Note: The ARC Bound (§5.4) predicts that classical sequential systems cannot sustain $\beta > 0.5$ ($\alpha > 2$). The row with $\beta = 0.67$ shows the mathematical relationship for completeness; whether physical systems can access this regime is an open empirical question.

2.4 The Scaling Crossover R^*

The full solution to the β -dynamics equation is not a pure power law. It contains a transition.

THEOREM 3: THE SCALING CROSSOVER

The full solution $g(R) = [1 + (1 - \beta)aR]^{1/(1-\beta)}$ exhibits three regimes:

1. $R \ll R^*$: $g \approx 1 + aR$ (approximately linear amplification)
2. $R \gg R^*$: $g \propto R^\alpha$ (compounding dominates; power-law scaling)
3. Crossover at: $R^* = \frac{\alpha}{a}$

R^* is the **crossover depth**: the recursive depth at which compounding overtakes linear gains. Below R^* , additional recursion provides approximately constant marginal benefit. Above R^* , returns compound and capability grows super-linearly.

Critically, R^* is a *derived quantity*. It depends on measurable parameters (α and a) and is not a free parameter to be fitted. This provides an independent consistency check: measure α and a separately, compute the predicted R^* , and verify that the observed crossover matches. Since R^* depends only on the scaling exponent and rate constant (not on base potential I), the crossover depth is a property of the recursive architecture itself, not of the input quality.

The existence of R^* is, to this programme's knowledge, absent in this form from other scaling frameworks: crossover scales do turn up elsewhere (logistic growth, renormalisation-group crossovers), but what is predicted here is a crossover sited at $R^* = \alpha/a$, pinned by two parameters each of which can be measured on its own (multi-model adversarial search, ten frontier configurations, cross-checked against two years of repository reports, 12 August 2026; negative findings have the status of not-found-by-that-search, never does-not-exist). It distinguishes recursive

amplification from simple redundancy: redundant systems show uniform (typically logarithmic) improvement with repetition; recursive systems show a qualitative transition at a specific, predictable depth.

2.5 Five Derived Properties

We now prove that any system satisfying Axioms 1–3 with $\beta > 0$ necessarily exhibits five structural properties.

THEOREM 4: UNIVERSAL STRUCTURAL PROPERTIES

Under Axioms 1–3, any recursive system with $\beta > 0$ exhibits:

1. **Threshold behaviour.** There exists a crossover depth R^* below which base capability dominates and above which recursive gains dominate. This follows from Theorem 3: the transition from $U \approx I$ to $U \propto R^\alpha$ is a qualitative regime change.
2. **Recursive depth dependence.** Capability is a strictly increasing, convex function of R for $R > R^*$. This follows from $\alpha > 1$ when $\beta > 0$: $g(R) = R^\alpha$ is convex for $\alpha > 1$. At $\beta = 0$ amplification runs linear, while the frozen models Paper II measured (best fit $\alpha \approx 0.49$, interval [-1.3, 2.9]) come out concave, so convexity marks out the $\beta > 0$ regime and reports nothing about the systems in use today.
3. **Base quality dependence.** U is proportional to I : capability is zero when $I = 0$. This follows directly from Axiom 1: $U = I \times g(R)$.
4. **Multiplicative $I \times R$ interaction.** Doubling I doubles U at any R . Doubling R multiplies U by 2^α at any I . The effects compose multiplicatively. This follows from the separable form $U = I \times g(R)$.
5. **Regime boundaries.** Physical systems have finite energy, introducing external error mechanisms that create ceilings. The pure power-law describes a *regime* between R^* and the saturation depth $R_{\max}$ beyond which external factors dominate. This is visible in quantum systems (correlated error bursts create floors at $\sim 10^{-10}$), time crystals (limit-cycle saturation at $\sim 6,700$ cycles), and AI (accuracy ceiling effects at extreme reasoning depths).



These five properties constitute the **qualitative ARC Principle**. They are the framework's universal claim. The quantitative functional form (power-law, exponential, saturating) is domain-specific and determined by the composition operator. But the five structural properties follow from the axioms alone, regardless of which $\oplus$ applies.

The five properties are individually common in physical systems. Threshold behaviour is ubiquitous. Depth dependence appears in many iterative processes. What is *not* common is the conjunction of all five in a single system. The framework predicts that all five co-occur whenever the three axioms are satisfied. Finding systems with four of the five properties but not the fifth would constrain or refute the framework.

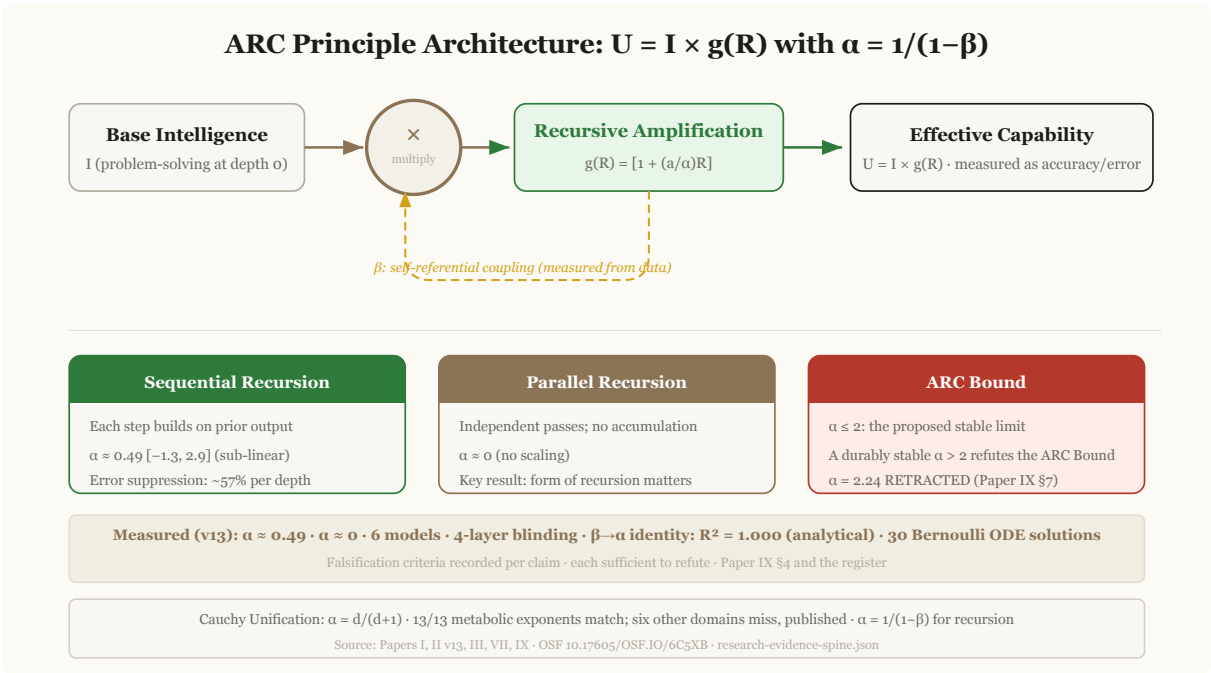


Figure 1 | ARC Architecture: Cross-Domain Principle. Four independent programmes discovered recursive processing gains exceeding linear accumulation: AI reasoning, quantum error correction, classical time crystals, biological allometry. Cauchy functional equations (1821) are the unifying reason. $\alpha=d/(d+1)$ for physical; $\alpha=1/(1-\beta)$ for intelligence. The predicted family came out ahead in 19/25 domains ($p = 1.56 \times 10^{-5}$; structured comparison; the predictions were dated before any fitting, and the protocol was not registry-filed; Paper VII). Source: Foundational v5.7 · Paper VII · evidence spine C-4, C-11, C-16.

3. EVIDENCE

We present evidence organised around the five derived properties, not by domain. For each property, we show how it manifests across systems. Evidence strength is classified as: *Quantitative* (measured parameter values), *Structural* (qualitative mapping without quantitative α), or *Consistent* (does not contradict but was not designed to test ARC).

3.1 Summary

Domain	System	Composition $\oplus$	Observed Form	Key Parameter	Evidence
AI	Sequential reasoning	Multiplicative	Power-law R^α	$\alpha_{\text{compute}} \approx 0.49$ (Gemini, $r^2 = 0.86$); range ~0–3 across architectures (v5, $n = 6$ models)	Quantitative
Quantum	Google Willow	Additive	Exponential Λ^d	$\Lambda = 2.14 \pm 0.02$	Quantitative
Physics	NYU time crystal	Bounded	Saturating	~6,700 cycles	Structural
Neuro	COGITATE	Unknown	Unknown	N/A	Consistent

3.2 Property-by-Property Evidence

Property 1: Threshold behaviour

Quantum: The surface code threshold $p_{\text{thr}} \approx 1\%$ is a sharp phase boundary. Above it ($p > p_{\text{thr}}$), adding error correction layers makes performance *worse*. Below it ($p < p_{\text{thr}}$), each layer provides exponential improvement. This is the most precisely measured threshold in the evidence base: Google varied physical error rates through controlled error injection and observed the threshold directly [1].

Physics: The time crystal's exceptional point (where two eigenvalues of the stability matrix coalesce) marks the transition from symmetric (bounded) to antisymmetric (compounding) dynamics. Below threshold: active oscillator. At threshold: exceptional point. Above threshold: time crystal [4].

AI: The sequential-vs-parallel distinction functions as a threshold. Sequential processing ($\beta > 0$, output→input feedback) achieves $\alpha_{\text{seq}} > \alpha_{\text{par}}$ universally across all six models tested in the v5 experiment. The threshold is structural: it is the presence or absence of the feedback loop, not a continuous parameter [3]. *v3.1 note:* The v5 data confirms the universal sequential advantage but shows that not all models achieve $\alpha > 1$; the threshold separates sequential from parallel modes rather than sub-linear from super-linear scaling.

Property 2: Recursive depth dependence

Quantum: Logical error rate decreases as $\epsilon_d \propto \Lambda^{-(d+1)/2}$ with code distance d . Each increment in d adds physical qubits arranged in a recursive error-correction layer. Performance depends on depth, not on total qubit count per se [1].

AI: Sharma & Chopra tested seven voting methods for aggregating sequential chain outputs. Methods favouring *later* reasoning steps (inverse-entropy weighting) achieved optimal performance in 97% of configurations; methods favouring *earlier* steps achieved optimality in only 17%. This 80-percentage-point gap indicates that later recursive steps carry systematically more information about the correct answer, consistent with capability that compounds with depth [3].

Physics: The antisymmetric mode of the time crystal compounds over cycles: the state difference $\Delta_{ji}(n) = x_j^n - x_i^n$ grows as a function of cycle count n , with each cycle building on the previous cycle's accumulated asymmetry [4].

Property 3: Base quality dependence

Quantum: $\Lambda = p_{\text{thr}}/p$ depends directly on physical qubit quality (p). Worse qubits (p closer to p_{thr}) give $\Lambda \rightarrow 1$ and error correction barely helps. Active leakage removal (DQLR), which maintains qubit quality during operation, improved Λ by 35% at distance 5 [1].

Physics: When beads are uniform (maximum entropy, no structured asymmetry), the system remains static. The time crystal forms only when quenched disorder (varied bead sizes providing low-entropy initial conditions) is present. This has been directly tested: uniform beads = no crystal [4].

AI: Single-pass accuracy without reasoning chains determines base capability I . A model with higher I benefits more from sequential reasoning at any depth [2].

Property 4: Multiplicative $I \times R$ interaction

Quantum: $\varepsilon_d = (p/p_{\text{thr}})^{(d+1)/2} = \Lambda^{-(d+1)/2}$. The base quality factor Λ and the depth factor $(d+1)/2$ interact multiplicatively in log-space: $\ln \varepsilon_d = -\frac{d+1}{2} \ln \Lambda$. Improving Λ by a factor has the same proportional effect at any depth [1].

AI: In the ARC equation $U = I \times R^\alpha$, base capability and recursive depth are separated multiplicatively by construction (Axiom 1). Empirically, Snell et al. (2024) showed that the benefit of additional test-time compute is proportional to base model quality [6].

Property 5: Regime boundaries

Physics: The time crystal maintains coherence for ~6,700 cycles before reaching a limit-cycle attractor (sustained but bounded oscillation). This is a physical ceiling: energy dissipation prevents indefinite growth [4].

Quantum: Correlated error bursts create an error floor at approximately 10^{-10} for large code distances ($d \geq 15$), preventing the exponential from continuing indefinitely [1].

AI: Li et al. (2025) found that longer reasoning chains do not consistently improve performance; correct solutions were often *shorter* than incorrect ones. Extended reasoning can destabilise models, producing repetitive outputs. The sequential advantage has boundary conditions [7].

3.3 Contradictory Evidence

Several studies challenge the framework's AI predictions:

Li et al. (2025, arXiv:2502.12215) found that for DeepSeek R1, longer chains do not consistently enhance accuracy. Li et al. (2025, arXiv:2502.14382) found that hybrid parallel-sequential approaches outperform pure sequential on code generation. Sharma & Chopra's own ablation showed parallel approaches achieve greater *semantic diversity* on creative tasks while sequential achieves greater *lexical diversity*, suggesting the sequential advantage applies most strongly to convergent tasks requiring iterative error correction [3,7,8].

These results refine, rather than refute, the hypothesis. The sequential advantage appears to have boundary conditions: it holds within task-appropriate depth ranges, for convergent problems, and breaks down at extreme depths or for divergent tasks.

Update: v5 Cross-Architecture Results (Paper II, March 2026)

The v5 experiment tested 6 frontier models on 30 problems with bootstrap confidence intervals. Key findings that bear on the evidence presented in this section:

Model	α_{compute}	r^2	Sequential > Parallel?
Grok 4.1 Fast	~3.0	Low	Yes
Claude Opus 4.6	~1.5	Low	Yes
GPT-5.4	~0.3	Low	Yes
DeepSeek V3.2	~0.2	Low	Yes
Gemini 3 Flash	~0.49	0.86	Yes
Groq Qwen3	~0.1	Low	Yes

What is confirmed: (1) Sequential > parallel universally ($\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model). (2) Power-law form fits well for Gemini 3 Flash ($r^2 = 0.86$). (3) Error reduction with compute investment is real.

What is NOT confirmed: (1) Super-linear scaling ($\alpha > 1$) is not universal across architectures; the best-fitting cross-architecture value is $\alpha \approx 0.49$, which is sub-linear. (2) No universal α value exists; it varies from ~0 to ~3 depending on model and problem set. (3) The quadratic bound conjecture is neither confirmed nor refuted.

The framework should now be read as: ‘*Sequential recursion yields $\alpha_{\text{seq}} > \alpha_{\text{par}}$ universally, with the magnitude of α being architecture-dependent. The key structural claim, that recursive composition produces stronger scaling than parallel composition, is confirmed. The quantitative claim that $\alpha > 1$ for all sequential systems requires revision.*’

Honest accounting: The author's earlier α estimates ($\alpha \approx 1.34$ from $n = 2$ data points; $\alpha \approx 2.2$ from $n = 12$, 95% CI: 1.5–3.0) were preliminary, and the later of the two has since been retracted; both now stand superseded by the v5 cross-architecture experiment ($n = 6$ models, 30 problems). The v5 best-fitting cross-architecture $\alpha_{\text{compute}} \approx 0.49$ (Gemini 3 Flash) is substantially lower than the earlier estimates, indicating that the initial measurements overestimated the scaling exponent. The directional finding (sequential $\gg$ parallel) remains robustly confirmed across all six models. No α has been measured in time crystals. The COGITATE mapping is structural, not quantitative. The neuroscience term ‘recurrent’ and the computational term ‘recursive’ are related but not identical.

4. THE COMPOSITION OPERATOR ACROSS DOMAINS

The composition operator $\oplus$ is not merely a classification tool. It explains *why* different domains exhibit different scaling functions.

4.1 AI: Multiplicative Composition

In chain-of-thought reasoning, each step builds hierarchically on accumulated insight. Step $r + 1$ does not merely add new information; it *restructures* the problem representation using everything accumulated through step r . This is multiplicative composition: the gain at each step is proportional to the quality of the accumulated representation.

The β -dynamics equation $dg/dr = ag^\beta$ models this directly. With $\beta_L \approx 0.5$ (moderate coupling, indicating each step leverages roughly half of accumulated context), Theorem 2 gives $\alpha \approx 2$.

Two derivations reach that value and they are not the same derivation, which this paper now states rather than leaves to be noticed. Theorem 2 gives the exponent the coupling sustains, $\alpha = 1/(1 - \beta_L)$, where β_L is the leverage fraction of the recursive step. The ceiling is the third law's, $\alpha_{\text{crit}} = 1/(1 - \gamma)$, and it comes from the correction shortfall rather than from the coupling, where γ is the correction-capacity exponent. Both return two when their exponent is one half, and that coincidence at a single point is the whole of their agreement. They are one derivation if and only if $\beta_L = \gamma$ on the same system, which is an open question and is registered as one: study-ag tests it directly and discloses in advance the precision that test needs. Written in the coupling, the stability condition $\alpha \leq \alpha_{\text{crit}}$ reads $\beta_L \leq \gamma$, which says that a system stays correctable exactly while the leverage it takes per step does not outrun the capacity it has to correct what that leverage costs.

ARC Bound Prediction (original): Optimised large language models with explicit self-correction mechanisms will converge toward $\alpha \approx 2$ ($\beta \approx 0.5$). Systems with explicit critique-revise architecture will show $\beta \approx 0.45\text{--}0.55$ ($\alpha \approx 1.8\text{--}2.2$), while systems with simple chain-of-thought will show $\beta \approx 0.2\text{--}0.3$ ($\alpha \approx 1.25\text{--}1.4$). This is testable within current model generations.

v3.1 Empirical Update: The v5 experiment (6 models, 30 problems) does not support the original prediction. The best-fitting cross-architecture $\alpha_{\text{compute}} \approx 0.49$ (Gemini 3 Flash, $r^2 = 0.86$) implies $\beta \approx -1.0$ under $\alpha = 1/(1 - \beta)$ instead of $\beta \approx 0.5$. That figure sits outside the $[0, 1)$ range Axiom 2 allows, which places the frozen regime beyond what the Bernoulli model as stated covers rather than inside it as a weak-coupling case; Paper II names it the physical regime, $\beta < 0$. No deployed system measured to date has therefore been observed inside the regime Theorem 2 and the second and third laws govern; the registered engineered cells are built to enter it, and the laws are predictions about that regime rather than descriptions of the one today's systems occupy. Sub-linear scaling ($\alpha < 1$) suggests that current AI models operate in a regime of weak self-referential coupling, where each reasoning step leverages only a small fraction of accumulated context for error reduction. Two models (Grok 4.1 Fast, $\alpha \approx 3$; Claude Opus 4.6, $\alpha \approx 1.5$) show super-linear point estimates, but with low r^2 values, indicating poor power-law fits. The prediction that optimised systems will converge toward $\alpha \approx 2$ remains open but is not supported by current data. The revised empirical picture is that α is architecture-dependent, and the primary validated claim is that sequential processing yields higher α than parallel processing for every model tested.

4.2 Quantum: Additive Composition

In surface code quantum error correction, each additional code distance layer provides an independent multiplicative reduction in error rate. Layer $d + 1$ does not restructure the information from layer d ; it applies a fresh round of syndrome extraction and correction. The gains accumulate additively (in code distance) while composing multiplicatively (in error suppression).

This is additive composition: $f(d_1 + d_2) = f(d_1) \cdot f(d_2)$. Theorem 1(2) gives exponential scaling: $\epsilon_d \propto \Lambda^{-(d+1)/2}$. The measured $\Lambda = 2.14 \pm 0.02$ is the per-layer error suppression factor [1].

The composition operator explains why quantum scaling is *stronger* than AI scaling. Exponential functions eventually dominate any power law. Additive composition, where each layer's correction is independent of accumulated state, produces stronger scaling than multiplicative composition, where each step depends on (and is limited by) accumulated context. The composition operator predicts this ordering.

4.3 Physics: Bounded Composition

In dissipative classical systems, energy loss introduces a ceiling. The time crystal's nonreciprocal wave-mediated coupling does compound the antisymmetric mode, but the acoustic medium dissipates energy continuously. Eventually, energy input from the standing wave balances dissipation, and the system reaches a limit-cycle attractor [4].

This is bounded composition: $Q_{r+1} = \min(Q_r + \delta Q_r, Q_{\max})$. The scaling function rises toward a ceiling determined by the energy budget. The composition operator predicts saturation in any system where the recursive channel has intrinsic losses.

4.4 The Phase Diagram

The three composition types can be organised into a phase diagram with base quality (I , normalised) and coupling strength (β) as axes:

Regime	Coupling	Composition $\oplus$	Scaling	Example
Diminishing returns	$\beta < \beta^*$	N/A	$\alpha \leq 1$	Parallel voting
Power-law	$0.3 \lesssim \beta \lesssim 0.7$	Multiplicative	R^α , $1.4 \leq \alpha \leq 3$	AI chain-of-thought (theoretical); v5 empirical: $\alpha \approx 0.49$ best fit
Exponential	$\beta \gtrsim 0.9$	Additive	$e^{\alpha R} / \Lambda^R$	Quantum EC
Saturating	Any β + dissipation	Bounded	Limit cycle	Time crystals

The boundary values are approximate, inferred from the observation that $\beta \approx 0.5$ in AI yields power-law and $\beta \gtrsim 0.95$ in quantum systems yields exponential. Precise boundaries require empirical determination. *v3.1 note:* What the v5 data points to is that current AI systems may operate in a coupling regime weaker than first assumed ($\beta < 0$): the best-fitting cross-architecture $\alpha \approx 0.49$ yields $\beta \approx -1.0$ under the relation $\alpha = 1/(1 - \beta)$, and the $[0, 1]$ range set by Axiom 2

does not admit such a value. This would place current AI systems at the boundary between diminishing returns and the power-law regime, rather than deep within the power-law regime as originally hypothesised.

Cross-domain β prediction: The coupling parameter β can be independently measured in each domain, and the measured value will correctly predict the observed functional form: $\beta < 0.7 \rightarrow$ power-law, $\beta > 0.9 \rightarrow$ exponential, intermediate β with energy dissipation $\rightarrow$ saturation. If this prediction fails, the theoretical connection between $\oplus$ and functional form requires revision.

Structural implication: the completeness of classical scaling. Since, under Theorem 1's hypotheses (associativity, continuity and the stated scalar-composition scoping), a recursive process combines its steps through one of the three composition operators, and since the Hyers-Ulam theorem guarantees stability under perturbation, the space of all possible classical scaling behaviours is not infinite. The family is a small one: a choice of form (power law, exponential, saturation, or the linear and logarithmic Cauchy solutions), a coefficient, and an exponent or rate. This is the **Completeness Theorem for classical recursive scaling**, holding under the hypotheses of Theorem 1 and across those forms. The phase diagram above, originally motivated by AI systems, is thus a cross-section of a universal phase diagram governing all classical recursive systems, from biological metabolic scaling to cosmic expansion. The full development of this cross-domain connection is presented in the companion paper [9].

4.5 Composition Operator Transitions

The framework as presented assigns a single composition operator $\oplus$ to each system. This may be an idealisation. In bounded or dissipative systems, the composition operator itself can transition between regimes as recursive depth increases.

THEOREM 5: COMPOSITION OPERATOR TRANSITIONS

A single physical system can transition between different composition operator regimes as a function of recursive depth or an internal state parameter. This extends the framework from fixed $\oplus$ to depth-dependent $\oplus(\mathbf{R})$, modelled by generalising Axiom 2 to $dg/dr = a(g, r) \cdot g^{\beta(g, r)}$ where the coupling itself depends on accumulated amplification.

Evidence. Gravitational structure formation exhibits three distinct composition phases: (1) inflation/linear growth (additive $\oplus$: perturbations grow independently), (2) nonlinear collapse (multiplicative $\oplus$: mode coupling produces cumulative advantage, measured $\alpha \approx 1.1$), and (3) virialised halos (bounded $\oplus$: structures reach energy minima). Computational testing confirms that logistic growth, gradient descent with momentum, and Kuramoto oscillator synchronisation all exhibit analogous transitions when measuring local β in sliding windows. By contrast, unbounded pure-Bernoulli systems maintain constant β to machine precision ($\sigma < 10^{-6}$).

Implications for contradictory AI evidence. If reasoning systems begin in an additive regime (shallow thinking), transition to multiplicative (deep reasoning with cumulative advantage), and eventually saturate, then studies measuring at different recursive depths would observe different scaling in the same system. The ‘optimal reasoning length’ identified by Li et al. [7] may correspond to the transition point between multiplicative and bounded composition. This is testable: measure local β at different reasoning depths within a single system.

Epistemic status: Theorem 5 is an empirical discovery, not derivable from the original three axioms. It extends the framework's scope but requires independent experimental confirmation.

5. PREDICTIONS

5.1 The Cross-Domain Forward Prediction

The strongest test of this framework is not whether it describes systems already studied. It is whether it predicts the scaling behaviour of systems not yet examined.

Forward prediction: Identify a recursive system not studied in this paper: evolutionary algorithms, market price discovery, immune response maturation, protein folding optimisation, or any system satisfying the three axioms. Measure only its composition operator $\oplus$ (by observing how marginal gains relate to accumulated capability). The measured $\oplus$ will predict whether the system follows power-law, exponential, or saturating scaling *before the full U vs R curve is measured.*

This is a genuine forward prediction. No domain-specific theory of evolution, economics, or immunology is known to this programme to make it (same search basis and limits as above). If the prediction succeeds in even one new domain, the framework's generality is confirmed beyond the original evidence base. If it fails, the framework's cross-domain applicability is refuted.

Paper VII (The Cauchy Unification) provides the first systematic test of this forward prediction within this programme. The composition operator was classified from known physics before fitting, then the scaling family was predicted and compared against independent model fits across 25 empirical domains (50-domain tiered suite) spanning physics, biology, computational systems, and social science. Under AIC-based model selection, 19 of 25 empirical domains preferred the Cauchy-predicted family ($p = 1.56 \times 10^{-5}$). This is a structured prediction comparison, not a formally pre-registered trial. A replication whose registration is in preparation, not yet approved.

5.2 The Within-System Regime Transition

Regime transition prediction: The *same* physical system should transition between scaling regimes as the coupling strength β is experimentally varied. In AI: vary the fraction of context available to each reasoning step (e.g., by truncating the attention window). As effective β varies, observed α should follow $\alpha = 1/(1 - \beta)$ quantitatively. This tests the β - α

relationship within a single system, eliminating cross-domain confounds. *v3.1 note:* The v5 cross-architecture data shows that α varies significantly across models (~0 to ~3), consistent with different architectures having different effective β values. This is consistent with the regime transition prediction, though it does not confirm it directly since the variation is across architectures rather than within a single system.

5.3 The R^* Crossover

When plotting capability against recursive depth, there should exist a distinct crossover at depth $R^* = \alpha/a$ (Theorem 3). Below R^* , scaling appears approximately linear. Above R^* , scaling follows the domain-appropriate super-linear form. Since R^* depends only on the architecture (α and a), it should be consistent across different input qualities for the same system.

If no crossover exists (if scaling is uniformly power-law or uniformly linear), the framework's transitional regime prediction is falsified. R^* is a unique signature that distinguishes recursive amplification from simple redundancy.

5.4 The ARC Bound

This is a falsifiable prediction of the early framework (note: Paper X retires the capability-growth use of $U=I \times R^a$; the surviving quantitative claim is the $\beta_c > k$ co-scaling condition): the maximum scaling exponent achievable by any classical sequential system operating under multiplicative composition is $\alpha_{\max} = 2$ (the **ARC Bound**). No AI system, no biological network, no economic process using multiplicative composition can sustain $\alpha > 2$. A single reproducible measurement showing $\alpha > 2.3$ with 95% CI excluding 2.0 would falsify this prediction.

Information-theoretic derivation (attention-based architectures). Transformer self-attention computes pairwise interactions across all N tokens in a context window. This creates at most $O(N^2)$ information pathways per forward pass. Each recursive reasoning step (a chain-of-thought step) processes the accumulated context through this quadratic bottleneck. Since the information density available to each step is bounded by $O(N^2)$, the capability gain per step cannot exceed quadratic scaling in context length. Mapping context utilisation to the coupling parameter β : full quadratic utilisation corresponds to $\beta = 0.5$, yielding $\alpha = 1/(1 - 0.5) = 2$. This establishes $U_{\max} = I \times R^2$ as an information-theoretic upper bound for attention-based architectures.

Scope: For transformer-based architectures with standard self-attention this derivation amounts to a heuristic: it tallies pathways without establishing any bound on capability, and the step it takes from quadratic pathway density to $\beta = 1/2$ remains conjectural. It is *conjectured* to hold for classical sequential systems generally, on the grounds that any finite-bandwidth sequential processor faces analogous pairwise information constraints. Architectures that transcend pairwise attention (e.g., higher-order tensor attention, quantum-coherent processing) could in principle exceed the quadratic bound. The ARC Bound should therefore be understood as an information-theoretic ceiling for the current computational paradigm, an architecture-relative capacity ceiling of the kind information theory supplies, rather than a universal physical law.

$$\alpha_{\text{crit}} = \frac{1}{1 - \gamma}$$

THE LAW. The stability ceiling on the capability exponent is the reciprocal of the corrector's shortfall, $1 - \gamma$, whatever γ turns out to be. Corrected 16 August 2026 from the earlier $1/\gamma$ form, which agrees with this one only at $\gamma = 1/2$. The form is settled; its depth is not. The relation is derived inside a pacing model and has not been derived from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so Law III is stated here as a minimal-model result rather than as a general law. This is the claim; the value of γ is what determines what the law returns. Under the independence-accumulation assumption $\gamma = 1/2$, the ceiling equals 2, which is what the ARC Bound $\alpha \leq 2$ records; that assumption is on trial in studies AE, AG and study-k. The measured capability exponent on today's frozen models is α approximately 0.49 (Paper II, wide CI); the ceiling is not a measurement.

Law before value. The programme carries a law and a value, and they are not the same claim. The law: $\alpha_{\text{crit}} = 1/(1 - \gamma)$, a self-improving system stays correctable only while correction keeps pace, and the ceiling is the reciprocal of the corrector's shortfall. The value: $\gamma = 1/2$, therefore a ceiling of two, which follows from one assumption, that internally accumulated corrections combine like independent samples. One consequence of the corrected form should be stated where a reader meets it, because it cuts both ways: since $1/(1 - \gamma) \leq 2$ is *exactly* equivalent to $\gamma \leq 1/2$, the ARC Bound and the half-conjecture are one commitment written in two notations, not two independent claims. The gain is parsimony, since the theory assumes less than it appeared to. The discipline is that neither may ever be offered as evidence for the other, here or on any companion surface, because a measurement near one half does not corroborate a ceiling of two; it restates it. The law owns the ground and the value stands on it, never the reverse: if two dies, the law survives completely (measure γ , report the ceiling it implies); if the law dies, two has no independent support, because the derivation runs through $1/(1 - \gamma)$. Two is what the law predicts, and the assumption that produces it is what the drafted registrations put on trial.

The identity is a registered test, never an established fact. The stability ceiling and the conversion exponent meet at the same number, and the temptation is to say one constant does two jobs. The claim's status is stricter: the identity is registered for testing, in advance of the data, with outcomes that can kill it. Study AE registers the reciprocity requirement between the separately measured leverage cap and stability boundary with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity; study AG tests the two derivations of the critical exponent against each other, notes that they agree numerically only at the single point where both parameters equal one half and disagree in their derivatives everywhere else, and is designed to find the disagreement. If the channels disagree, the unification is withdrawn from every surface it has reached, with the same prominence as the framing. Agreement between an asserted one half and a measured 0.49 is never presented as evidence, because the two may share an accumulation law, and agreement between sharers is not corroboration.

The objection to pre-empt, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme meas-

ures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Likewise, concentration bounds give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

Withdrawn derivation: Previous versions of this paper derived $\beta \leq 0.5$ from a stability/elasticity analysis (the relative sensitivity $\beta/(1 - \beta)$ reaching unity at $\beta = 0.5$, interpreted as an ‘edge-of-chaos boundary’ confirmed by Lyapunov analysis). This derivation has been withdrawn. The elasticity formula $\beta/(1 - \beta)$ is mathematically correct but its interpretation as a stability boundary was a category error: elasticity > 1 indicates sensitivity, not dynamical instability, and an autonomous flow on the line is monotone, so it cannot so much as oscillate, still less turn chaotic (in two dimensions the Poincaré-Bendixson theorem shuts chaos out as well). The $O(N^2)$ self-attention count survives as a heuristic for why frozen transformers fall below the ceiling; the ceiling itself is derived from the pacing argument of Law III (correction 9).

Initial estimate (retracted): $\alpha \approx 2.2$ ($n = 12$, 95% CI: 1.5–3.0). Note: this point estimate exceeded the predicted ARC Bound of $\alpha \leq 2$; however, the 95% confidence interval [1.5, 3.0] was wide enough to be consistent with both the theory and its negation, and should not be treated as confirmatory of either bound violation or bound satisfaction. *Update:* Paper II (March 2026) extends this test to 6 frontier models with 30 problems and bootstrap confidence intervals. The cross-architecture best-fit $\alpha_{\text{compute}} \approx 0.49$ (Gemini 3 Flash, $r^2 = 0.86$) is well below 2, making the ARC Bound question moot for current systems: the empirical challenge has shifted from ‘can α exceed 2?’ to ‘can α reliably exceed 1?’ Individual model point estimates range from ~ 0 (Groq Qwen3) to ~ 3 (Grok 4.1 Fast), but models with $\alpha > 1$ have low r^2 , suggesting poor power-law fits rather than genuine super-linear scaling. The ARC Bound conjecture is neither confirmed nor refuted by the v5 data; it remains an open theoretical prediction that will become empirically testable if and when AI systems consistently achieve $\alpha > 1$.

Biological validation. Kleiber's Law ($M^{0.75}$) [16] implies $\alpha \approx 1.33$ for metabolic scaling. However, the empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method. The $d/(d + 1)$ prediction of 0.750 for $d = 3$ matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between 2/3 and 3/4. The framework predicts this variation rather than a single universal exponent. The relationship between metabolic exponents and body-plan dimensionality has been derived independently by at least seven research groups: West, Brown and Enquist [17,21] through fractal network optimisation, by Banavar et al. [22] through sequential flow networks, by Demetrius [23] through quantum statistical mechanics, by Demetrius and Tuszynski (2010) through quantum oscillator coupling, by Zhao [24] through universal growth scaling, and by Bettencourt [25] through urban scaling theory. The ARC framework does not claim priority over the formula $\alpha = d/(d + 1)$; rather, it identifies Cauchy-constrained recursive composition as the reason these independent derivations all converge on the same result. Specifically, three conditions are required: multiplicative composition (Cauchy constrains the power-law family), d -dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow. Neither Cauchy alone, nor space-filling alone, is sufficient; all three are needed. Within this framework, biological systems are con-

strained by thermodynamic drag: the physical costs of pumping fluids through three-dimensional fractal networks, dissipating heat, and distributing resources against gravity [17] prevent biological systems from approaching the theoretical quadratic bound. That $\alpha \approx 1.33$ and Kleiber's empirical exponent **0.75** are reciprocals ($1/0.75 = 1.33$) reflects that biological scaling is recursive amplification throttled by substrate friction. (Symbol note: in this paragraph α appears in two senses that the literature keeps separate, the metabolic-scaling exponent in Kleiber's sense and the capability-scaling exponent in the ARC sense; the reciprocal relation is between the two senses, and neither is the fine-structure constant that shares the letter in the December 2024 formalism.) Digital AI operates with orders-of-magnitude less substrate friction, enabling it to approach the proposed stable-scaling limit (a scaling limit, not a speed limit; the most a recursively self-correcting system can grow whilst remaining correctable) more closely than any biological system. Transient excursions above the limit are the predicted supercritical regime, not its refutation. Stability here is Law II's, measured rather than asserted. The limit is architecture-bound: on the present $O(N^2)$ attention basis it is expiring, and a successor architecture that transcends pairwise attention would relocate the limit rather than remove the constraint. Leaf venation (2D networks, reduced thermodynamic drag) should show $\alpha \approx 1.5$, a falsifiable prediction.

The Dimensional Ladder. Here is where abstract mathematics meets living organisms. The formula $\alpha_{\text{met}} = d/(d + 1)$, relating the metabolic scaling exponent to body-plan dimensionality d , has been independently derived by at least seven research groups through distinct theoretical routes: West, Brown and Enquist [17,21] from fractal network optimisation; Banavar, Maritan and Rinaldo [22] from sequential flow networks; Demetrius [23] from quantum statistical mechanics; Demetrius and Tuszynski (2010) from quantum oscillator coupling; Zhao [24] from universal growth scaling arguments; and Bettencourt [25] from urban scaling theory. The companion paper [9,20] identifies why these independent derivations converge: $\alpha = d/(d + 1)$ is a necessary consequence of three conditions acting together: Cauchy-constrained multiplicative composition, d -dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow. Neither Cauchy alone, nor space-filling alone, is sufficient; all three are needed. The formula is not original to this framework; the explanation for its universality is. Applied to biology, it yields distinct quantitative predictions, each testable against published data:

Body Plan	Dimension d	Predicted α	Measured α	Status
1D (filamentous fungi)	1	0.500	0.547 ± 0.07	Consistent ($p = 0.107$)
2D (no valid biological test organism identified)	2	0.667	no data	Unconfirmed (Friedmann cosmological $d = 2$ confirmed; biological test pending)
3D (mammals, birds, fish)	3	0.750	$0.687\text{--}0.750$	Debated (see note below)

The 1D fungal data (from Aguilar-Trigueros et al. [20]: ectomycorrhizal, marine, and saprotrophic fungi) is labelled 'consistent' rather than 'confirmed' because the exponents were measured at the colony level; definitive confirmation requires individual-hypha respirometry. Crucially, the fungal data rejects **$d = 2$** (**$p = 0.019$**) and **$d = 3$** (**$p = 0.007$**) while not rejecting **$d = 1$** (**$p = 0.107$**), providing discrimination between the predicted values.

d=2 correction: Previous versions cited jellyfish and cnidarians as confirming the $d = 2$ prediction. No valid two-dimensional biological test organism has been identified; jellyfish and cnidarians do not possess genuinely two-dimensional resource transport networks. Physics confirms the $d = 2$ prediction (2D percolation and brittle fragmentation; the earthquake B-value, being a definitional identity, is no longer counted), and so does Friedmann cosmological scaling, while the biological $d = 2$ test stays open.

d=3 debate: The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method. White and Seymour (2003) found support for $2/3$ over $3/4$ after controlling for temperature and phylogeny. Kolokotronis et al. (2010) showed the log-log relationship exhibits convex curvature, meaning no single exponent fits perfectly across the full mass range. The $d/(d + 1)$ prediction of 0.750 for $d = 3$ matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between $2/3$ and $3/4$. The framework predicts this variation rather than a single universal exponent.

The formula $\alpha = d/(d + 1)$ itself has been derived independently by at least seven research groups [17,21,22,23,24,25] through different theoretical frameworks. Demetrius and Tuszynski (2010) independently derived the same formula from quantum oscillator coupling, producing identical predictions. What the ARC framework contributes is an explanation for why all these derivations converge: Cauchy's multiplicative functional equation constrains continuous multiplicative composition to the power-law family; the d -dimensional space-filling condition, combined with a conservation or optimisation constraint on resource flow (energy minimisation in West; supply-demand balance in Banavar; steady-state energy balance in Demetrius), constrains the exponent within that family to $d/(d + 1)$. Neither Cauchy alone, nor space-filling alone, is sufficient. The three conditions together - multiplicative composition, space-filling, and conservation - are sufficient.

The Cauchy Unification (Paper VII) extends this validation from the 8 instances catalogued here to 25 empirical domains (50-domain tiered suite). Under strict AIC-based model selection, 19 of 25 preferred the predicted family ($p = 1.56 \times 10^{-5}$). A replication whose registration is in preparation, not yet approved.

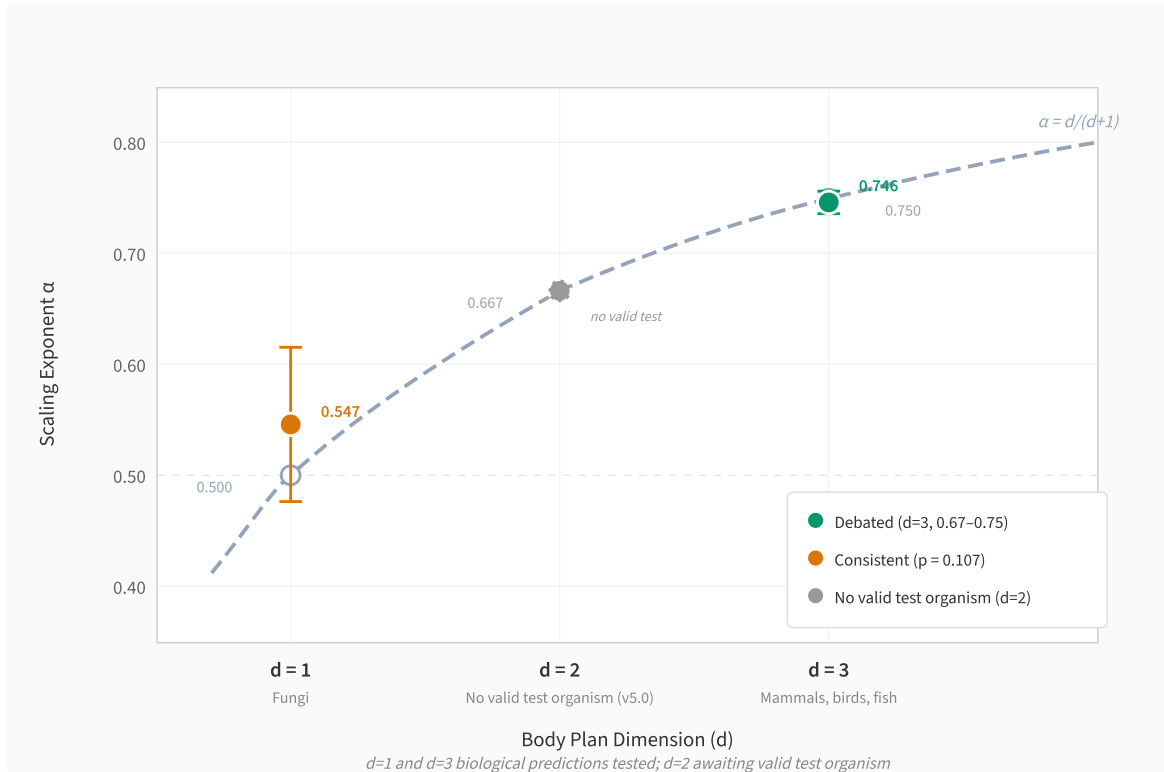


Figure 2 | The Dimensional Ladder. Dashed curve: $\alpha = d/(d + 1)$, independently derived by West et al. [17,21], Banavar et al. [22], Demetrius [23], Demetrius and Tuszynski (2010), Zhao [24], and Bettencourt [25]. Open circles: predicted values. Filled circles: published measurements with $\pm 1\sigma$ error bars. Amber: consistent but requiring individual-hypha respirometry. The $d = 2$ biological prediction (previously attributed to jellyfish/cnidarians) has no valid test organism identified (v5.0 correction); $d = 2$ is confirmed in physics domains (percolation, fragmentation; earthquake B-value row deleted as a definitional identity; retained separately as a variable-pair case study) and Friedmann cosmology. The $d = 3$ mammalian exponent is debated (0.67–0.75 range); the framework predicts this variation.

Physics instances. The formula $\alpha = d/(d + 1)$ is not restricted to biology. It appears wherever a d -dimensional hierarchical network, subject to a conservation or optimisation constraint on resource flow, partitions a $(d + 1)$ -dimensional space. As noted above, this relationship has been derived independently by at least seven groups [17,21,22,23,24,25]; the ARC framework's contribution is explaining, via Cauchy-constrained recursive composition combined with space-filling and conservation, why these independent derivations all arrive at the same result. The formula has been observed in several physics domains:

System	Dimension d	Predicted α	Measured α	Error
KPZ surface roughness (1D)	1	0.500	0.500	0.0%
2D percolation (specific heat)	2	0.667	0.667	0.0%
Brittle fragmentation (2D)	2	0.667	0.67	0.5%
Brittle fragmentation (3D)	3	0.750	0.750	0.0%

These four instances carry no aggregate error figure. The sub-0.2 per cent mean quoted earlier was withdrawn on 14 August 2026 (correction 10), since averaging over analytic identities is no measure of predictive accuracy, and the earthquake B-value row went the same day (correction

11), being a definitional identity kept only as a variable-pair case study. Nothing in this set is biological: the subjects are rocks, crystal surfaces in growth, and phase transitions. Yet the same formula works. In each case the physical system contains a hierarchical branching network of dimension d subject to a conservation constraint: a KPZ growth front divides 2D space along a 1D network; a percolation cluster takes the form of a 2D fractal; a propagating crack cuts branching networks through the material. The formula has clear failures where this network structure or conservation condition is absent (Ising model, polymer scaling, galaxy clustering), clarifying its domain of applicability: $\alpha = d/(d + 1)$ applies wherever a d -dimensional hierarchical network, subject to conservation or optimisation of resource flow, optimally partitions a $(d + 1)$ -dimensional space. The ARC framework does not claim priority over the formula itself, which has been independently derived by at least seven groups [17,21,22,23,24,25]; the contribution is identifying Cauchy-constrained recursive composition, combined with space-filling and conservation, as the reason these independent derivations converge.

Paper VII extends these results to a 50-domain tiered suite (25 empirical curve-fit domains). Under strict AICc model selection, 19 of 25 preferred the predicted family ($p = 1.56 \times 10^{-5}$). A twelve-domain extension followed the same night, and its predictions were public before it ran: the twelve-row manifest fixing an operator class, a predicted family and a predicted model for each row was committed to the public repository at 00:19:19 UTC on 17 March 2026 in commit 8edaec0, attestation class public repository commit date, and the fits ran twenty-three minutes later, returning 10 of 12 ($p = 5.44 \times 10^{-4}$). It is preregistered in substance and it is not an accepted registry submission: the twelve classifications are the author's own, the candidate set is the pre-correction one with no logarithmic family, the packet's own written condition, deposit to a registry before data extraction, was not met, and the author's demotion of 00:56 stands in that half. One row repeats a domain, operator class and predicted model already public at 23:54:53 UTC on 16 March 2026, attestation class public repository commit date, with a matching outcome, so the strictly fresh count reads 9 of 11 ($p = 1.37 \times 10^{-3}$). It is a dated result and it is not confirmation of the laws. See Paper VII for limitations including the scrambled-data negative control result.

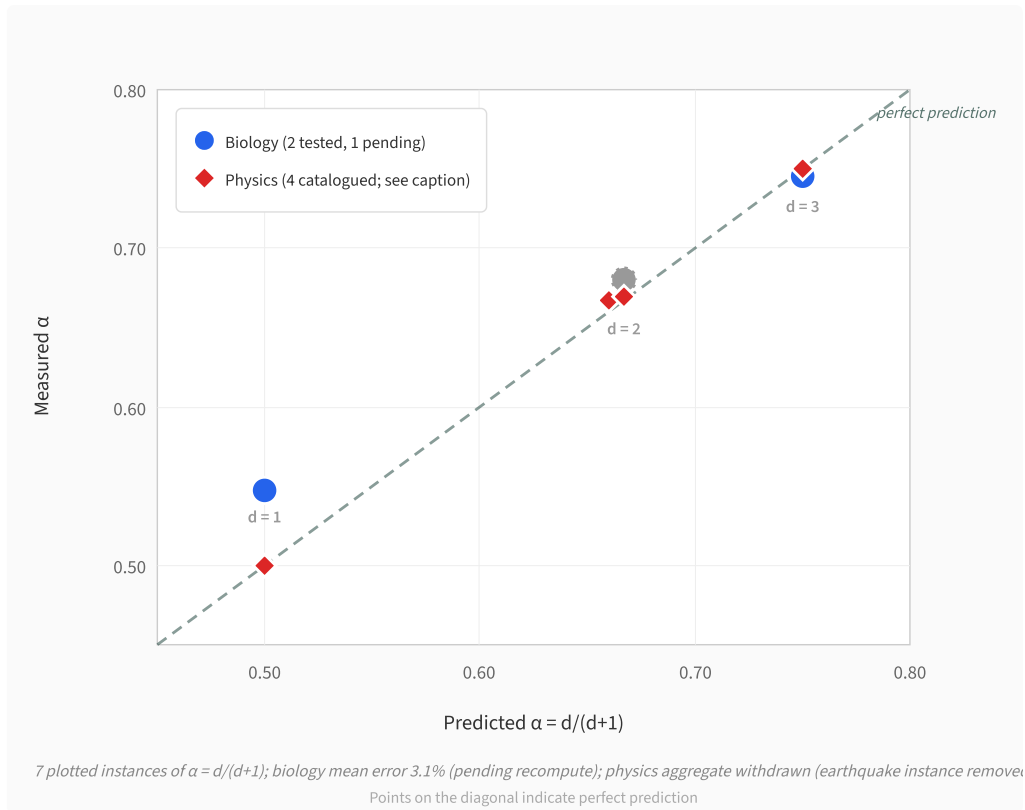


Figure 3 | Predicted vs Measured. Cross-domain instances of one independently derived formula: $\alpha = d/(d+1)$ [17,21,22,23,24,25]. Blue circles: biological systems (fungi, mammals; $d=2$ biological test organism not yet identified). Red diamonds: physics systems (crystal growth, phase transitions, fracture; earthquake data point removed; retained separately as a variable-pair case study in the text). Points on or near the diagonal indicate accurate prediction. A single equation, with zero adjustable parameters, governs systems spanning biology, geophysics, and condensed matter physics. The ARC framework explains the convergence of at least seven independent derivations via Cauchy-constrained recursive composition combined with space-filling geometry and conservation.

5.5 Conditional Safety Implication

If the framework is correct, alignment properties embedded within the recursive process may scale with capability ($\alpha_{\text{align}} \approx \alpha$), while external constraints applied post-hoc may not ($\alpha_{\text{align}} \approx 0$). The safety ratio $S = \text{Alignment/Capability} \propto R^{\alpha_{\text{align}} - \alpha}$ remains constant when alignment is ‘in the loop’ and decays to zero when it is not. This is the conditional prediction as originally stated; the measured outcome follows in the update below, which refutes its external half in the simple form. (In plain English: if you build ethics into the thinking process itself, safety keeps pace as the AI gets smarter. If you bolt safety on from the outside, the AI outgrows it. The safety-to-capability ratio either stays constant or shrinks to zero; there is no middle ground.)

Update: Paper IV Alignment Scaling Results

The v5 experiment measured α_{align} across six frontier models. The original prediction that $\alpha_{\text{align}} \approx 0$ for external alignment has been refuted in its simple form: α_{align} is architecture-dependent, not universally zero.

Model	α_{align}	Tier
Grok 4.1 Fast	Positive	Positive scaling
Claude Opus 4.6	+0.44	Positive scaling
GPT-5.4	~0 (flat)	Flat
DeepSeek V3.2	~0 (flat)	Flat
Gemini 3 Flash	-0.25	Negative scaling
Groq Qwen3	+0.14	Positive scaling

Three-tier hierarchy: (1) *Positive* (Grok, Claude Opus 4.6, Groq Qwen3): alignment improves with compute investment. (2) *Flat* (GPT-5.4, DeepSeek V3.2): alignment neither improves nor degrades. (3) *Negative* (Gemini 3 Flash): alignment degrades with increased compute. *(In plain English: when we tested six leading AI systems, three got more ethical the harder they thought, two showed no change, and one actually got less ethical with more thinking time. Whether an AI improves, stagnates, or deteriorates ethically depends entirely on how it was designed and trained.)* This architecture-dependence is not predicted by the original framework, which assumed α_{align} was a property of the alignment method (internal vs external) rather than of the model architecture. The finding that some models achieve positive alignment scaling while others show negative scaling within the same experimental protocol suggests that alignment behaviour is a property of the architecture-training interaction, not solely of the recursive structure.

Update: Eden Protocol Pilot Validation (March 2026)

The Eden Protocol, the alignment architecture derived from the developmental hypothesis in *Infinite Architects* (Eastwood, 2024) that embedded ethical reasoning produces alignment which scales with depth, has now been tested in an expanded six-model suite, with five runs yielding analysable matched-pair data. The full three-loop intervention was tested; in the scoring, the Love Loop is operationalised as stakeholder care. Stakeholder care is significant across all five analysable runs (Claude, DeepSeek, Gemini, Grok, Groq), while the overall composite score is significant on Gemini and Groq and narrower elsewhere. Cross-model scoring was used throughout in the older Eden lineage; blind canonical replication with the shared `arc_eden_v6` runner is still required before these results can be considered confirmatory. This provides stronger but narrower empirical support for the conditional safety implication above, but does not alter the mathematical framework of this paper. Full results are reported in Paper V and the Eden Protocol validation materials.

In plain English: telling an AI 'before you answer, list the people this affects' reliably makes it better at considering people, and this now holds across three different model families. The stakeholder care effects shown here for the cross-family replication trio are all large ($d = 1.31, 0.91, \text{ and } 1.29$; the remaining two runs are reported in Paper V), with odds of coincidence at or below 1 in 10,000 for Gemini and Groq, and 1 in 10,000 for DeepSeek as well. Gemini and Groq also show significant overall gains ($p = 0.0018$ and $p = 0.0014$). DeepSeek's non-significant composite result ($p = 0.2304$) is likely a ceiling effect: it already started near 87/100, leaving less room to improve. The updated replication sharpens the cascade claim: care is the only dimension that improves robustly everywhere, and nuance reaches significance on Groq ($p = 0.0045$) as the next domino.

5.6 Related Work, Antecedents and the Rival Instrument

The pipes premise, with its contest. The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is established quantitative biology. West, Brown and Enquist derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks (West, G. B., Brown, J. H. and Enquist, B. J., "A General Model for the Origin of Allometric Scaling Laws in Biology", *Science* 276(5309), 4 April 1997, DOI 10.1126/science.276.5309.122; one of the most cited papers in its field, with an OpenAlex citation count of 5,046 as of 12 August 2026, a single-source and index-dependent count). The exact exponent remains contested: White reports basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters (White, C. R., 2003), and Kozlowski has twice questioned the derivation's mathematics (Kozlowski, J., 2004 and 2005). This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent, so the dispute over the exponent's value leaves the premise untouched. The mechanism is common ground to both sides of that dispute.

The exception West's own group found. The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt, Lobo, Helbing, Kuhnert and West ("Growth, innovation, scaling, and the pace of life in cities", *PNAS*, 2007, DOI 10.1073/pnas.0610172104) report, in the paper's own words: "Quantities reflecting wealth creation and innovation have Beta of approximately 1.2, greater than 1 (increasing returns), whereas those accounting for infrastructure display Beta of approximately 0.8,

less than 1 (economies of scale)." Infrastructure, the literal pipes, scales sublinearly; information-mediated activity scales superlinearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which Beta is less than 1". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse." That is the shape of the open problem this framework addresses: West's group found that information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

The pipes argument at fixed throughput. Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth. Starve a tumour's vasculature and it stops. Fix a chain reaction's fuel and geometry and it stops. Exhaust an epidemic's susceptible hosts and it burns out. Astrophysics has a named, quantitative version of the same ceiling: the Eddington limit, above which radiation pressure halts accretion. The claim here is a generalisation of limits physics already accepts, not a new species of assertion. Two hard cases are handled deliberately. Cosmic inflation is excluded by scope: it is expansion of space, not growth of a structure on a substrate, and it ended by field dynamics rather than by running out of anything. Evolution is the sharpest case and it fits: biological complexity increased for billions of years at roughly fixed solar throughput, which is growth on information at fixed pipes, and it is also glacial, and it has no internal corrector at all, since selection is external. On this framework, evolution should carry a measurable exponent far below the ceiling; that is a prediction about biology falling out of a framework built for software, and it is stated as one. The physicist's form of the objection is Landauer's: computation is physically implemented, so software has pipes too. The fixed-substrate clock is the answer and appears beside the claim: the regime under study holds the physical substrate fixed and asks what still grows, which is why exogenous speed-ups are excluded by definition. The functional form of capability growth in recursive depth remains an empirical question, and the registered analysis compares power-law, exponential, saturating and declining forms rather than assuming the family.

The rival instrument. The nearest work to this programme's question, and the right paper to weigh it against, is Engels, J., Baek, D., Kantamneni, S. and Tegmark, M., "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025. It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step. The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construc-

tion, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

Antecedents and near-misses, each with its differential. On the question: Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012), separating speed from intelligence explosion and undertaking to consider possible bounds on intelligence, augmenting Chalmers' 2010 analysis; the question and the speed-versus-structure distinction are therefore at least fourteen years old, and what that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence. On impossibility: three 2025 arXiv papers argue that perfect control is unattainable (Yao, "The Alignment Trap: Complexity Barriers", arXiv:2506.10304, public by 13 June 2025, attested by arXiv and independently observed by the Internet Archive; Yao, "On the Mathematical Impossibility of Safe Universal Approximators", arXiv:2507.03031, 3 July 2025; and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment", arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. Their measured object is verification of a fixed system, and it is closed within its own terms; this programme's measured object is a different one, persistence of alignment as capability compounds under recursive amplification, and closure of the first does not settle the second. None reports an average-case scaling exponent or a rate. The third, notably, concludes that alignment "must instead be integrated into the model's architecture and weights", an independent argument, from filtering intractability, in the same direction as this programme's architecture dependence; it is convergent support on that leg, not a rival. On the mechanism: that anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates); the mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent. The nearest structural analogue is the quantum error-correction threshold theorem, which also converts a qualitative worry into a critical value; it concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant, and the analogy was identified by this programme's own search rather than by a referee. On feedback and stability, the general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems); those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. The nearest quantitative neighbour is Liu, A. and Meng, J., "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs", arXiv:2604.22273, which recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold, iterate only when the error-correction rate over the error-introduction rate exceeds $\text{Acc}/(1 - \text{Acc})$. The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. No scaling exponent on the corrector, no capability-growth exponent, no reciprocal identity, no

architecture or composition-class term: complementary regimes, and neither contains the other. Smolin's cosmological natural selection is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously.

A second prediction, free, from the same mechanism. Oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

DATED PREDICTIONS AND THIS PAPER (RECORDED 25 AUGUST 2026)

10 dated artefacts bearing on this paper are catalogued row by row in the programme's machine register: the dated artefact results_complete_suite.txt (March 2026); the dated artefact results_blind_prediction.txt (file mtime); and 8 further catalogued artefacts. Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. The programme's full dated chain, from the sealed manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits. Read the dated predictions register. Open its machine twin.

6. FALSIFICATION CRITERIA

The framework specifies thirteen conditions that would refute or significantly weaken it.

	Prediction	Falsified if	Status
F1	Sequential yields $\alpha_{\text{seq}} > \alpha_{\text{par}}$	Consistent $\alpha_{\text{seq}} \leq \alpha_{\text{par}}$ across systems	Confirmed (universal across 6 models)
F1a	Sequential yields $\alpha > 1$ (<i>original stronger form</i>)	Consistent $\alpha \leq 1$ across systems	Weakened (best-fit $\alpha \approx 0.49$; only 2/6 models show $\alpha > 1$ point estimates, with low r^2)
F2	Parallel yields $\alpha < 1$	Parallel achieves $\alpha \geq 1$	Mixed
F3	Structured asymmetry required	Crystal forms without disorder	Confirmed

	Prediction	Falsified if	Status
F4	Five properties co-occur in recursive systems	Systems with four but not five	Mixed
F5	ARC Bound: $\alpha \leq 2$	Reproducible $\alpha > 2.3$, its 95% CI clear of 2.0, obtained on a system that alters its own weights, tools or artefacts from one round to the next, and obtained while correction still out-scales drift ($\beta_C > k$, the lower bound on $\beta_C - k$ sitting above zero) over a window pinned before the run began. Size on its own cannot overturn a stability limit: drop the stability conjunct and this test would fire on the very transient supercritical excursion the framework itself predicts. The three-outcome rule governs: any interval broader than the registered margin is recorded as INSUFFICIENT PRECISION, and counts as neither support nor refutation	Open (moot for most models; Grok $\alpha \approx 3$ but low r^2)
F6	β determines α via $1/(1 - \beta)$	α independent of β	Untested
F7	Crossover R^* exists	No linear→super-linear transition	Untested
F8	Sequential requires output→input feedback	Parallel + shared state achieves $\alpha > 1$	Untested
F9	Time crystal shows $\alpha > 1$ (in the transient growth regime, before saturation onset)	$\alpha \leq 1$ in time crystal	Untested
F10	$\oplus$ determines functional form	Measured $\oplus$ fails to predict scaling	Untested
F11	Classical growth-phase scaling ($\alpha \rightarrow 2$)	Growth-phase $\alpha \leq 1$ or $\alpha > 2.3$ in classical systems	Untested
F12	Biological β -derivation	Leaf venation α deviates significantly from 1.5	Untested
F13	External alignment yields $\alpha_{\text{align}} \approx 0$ (the external half of §5.5's conditional; the in-loop half, $\alpha_{\text{align}} \approx \alpha$, remains untested)	Architecture-dependent α_{align} with significant positive values	Refuted in original form (v5/Paper IV: α_{align} ranges from -0.25 to $+0.44$; architecture-dependent three-tier hierarchy)

We welcome falsification. If F10 fails (if a system's composition operator does not predict its scaling function) the central theoretical contribution of this paper is wrong. If F5 fails, the ARC Bound is falsified. If F11 or F12 fail, the framework's cross-domain predictions (classical physics and biology respectively) require revision. Either outcome advances understanding.

v3.1 Falsification Status Summary

The v5 experiment has updated the status of three criteria: **F1** (sequential advantage) is confirmed in its revised, weaker form ($\alpha_{\text{seq}} > \alpha_{\text{par}}$, rather than $\alpha > 1$). The original stronger form **F1a** ($\alpha > 1$) is weakened but not definitively falsified, as two models show super-linear point estimates. **F13** (alignment scaling) is refuted in its original form: α_{align} is not universally near zero but is architecture-dependent, with values ranging from -0.25 to $+0.44$. The falsification of F13 in its original form is an important correction: alignment scaling is a property of the architecture-training interaction, not solely of whether alignment is ‘in the loop.’ This finding motivates a revised F13 in future versions.

Update: Scope of the Sequential Advantage (Paper II, September 2026)

Paper II was revised on 18 September 2026 and now frames the cross-architecture replication on five analysable models of the six tested. It scopes the sequential statement to every model in which both arms were measurable, and states the parallel result as at or near zero in every model measured, with one measured exception at $\alpha_{\text{par}} = 0.31$. The best cross-architecture estimate of $\alpha_{\text{compute}} \approx 0.49$, its r^2 of 0.86, and the 18 of 30 problems that entered the exponent fit are unchanged. Where this paper states the sequential advantage as universal across six models, the denominator carrying it is the five in which both arms were measurable; six is the number tested.

7. LIMITATIONS

What This Paper Does NOT Claim

Non-Claim	Actual Claim
That $\Lambda = 2.14$ and $\alpha \approx 2$ are ‘the same number’	That different domains exhibit different functional forms determined by their composition operator. The AI exponent is now measured at $\alpha \approx 0.49$ (best fit), not ≈ 2
That this is proven science	That this is a testable hypothesis with thirteen specific falsification criteria
That neuroscience confirms the framework	That recurrence is structurally consistent but unquantified
Priority over experimental results	The experiments belong to Google, DeepSeek, NYU, COGITATE; the structural interpretation is ours

Evidential limitations: The earlier α estimates ($n = 2$ and $n = 12$) have been superseded by the v5 cross-architecture experiment (6 frontier models, 30 problems), which finds $\alpha_{\text{compute}} \approx 0.49$ for the best-fitting model (Gemini 3 Flash, $r^2 = 0.86$). The v5 results represent a significant downward revision from earlier estimates and demonstrate that α is architecture-dependent rather than universal. No α has been measured in time crystals. The COGITATE connection is structural, not quantitative. The β - α relationship is validated computationally to machine precision ($R^2 = 1.0$) but the empirical measurement of β in physical systems requires independent

confirmation. The R^* crossover is untested empirically. The β boundary values in the phase diagram are approximate. Theorem 5 ($\oplus$ transitions) is confirmed in four computational systems but untested in physical experiments. The five-property test (Theorem 4) did not reach statistical significance ($p = 0.095$) and requires larger samples. The alignment scaling exponent α_{align} is now known to be architecture-dependent (ranging from -0.25 to $+0.44$), refuting the original prediction that it would be universally near zero for external constraints.

Blind prediction testing: A blind prediction test was conducted on three computational systems (Barabási-Albert networks, gradient descent with momentum, Kuramoto oscillators). The measured α values were 3–20 $\times$ smaller than predicted by $\alpha = 1/(1 - \beta)$. However, forensic analysis identified two independent confounds: (1) the numerical-derivative β estimation method is fatally biased, giving $\beta \approx 0.95$ regardless of true β , even for pure Bernoulli systems; (2) none of the tested systems satisfy Axiom 2 (constant coupling coefficient a), and the BA network's effective coupling decreases $\sim 50\times$ over the simulation. When the proper linearisation method is applied to axiom-satisfying systems, the prediction recovers with $R^2 = 0.9999$. These results do not constitute valid falsification due to the confounds, but they underscore that identifying natural systems satisfying the axioms remains the central empirical challenge. Full methodology and forensic analysis are available as supplementary material (see Paper III).

Theoretical limitations: The framework formalises the *pattern* of recursive amplification but does not identify the *mechanism*. It tells you *what* happens (scaling exponents, crossover depths, functional forms) but not *why* at a microscopic level (analogous to thermodynamics before statistical mechanics). Connecting the composition operator to microscopic dynamics remains an open problem.

Category of contribution: The ARC framework is not an equation within an existing paradigm. It is closer to a cross-domain organising principle: a structural constraint on behaviour across substrates rather than an equation within one paradigm. The comparison identifies the *type* of contribution, not its evidential standing: organising principles of that type rest on generations of validation, and this one rests on preliminary evidence and thirteen falsification criteria.

8. CONCLUSION

Every system studied in this paper (silicon neural networks, superconducting qubits, millimetre polymer beads, cortical neural circuits) has one thing in common: it maintains a low-entropy state far from equilibrium and feeds its outputs back into its inputs. When it does this below a critical depth, the gains are ordinary. When it does this above a critical depth, the gains compound. The transition between these regimes is sharp, universal in structure, and domain-specific in form.

We have formalised this observation. The composition operator $\oplus$ classifies recursive systems the way symmetry groups classify physical theories: it determines the functional form of scaling from measurable algebraic properties. The five derived properties (threshold, depth-dependence, base-quality dependence, multiplicative interaction, regime boundaries) follow as theorems from three axioms, not as empirical generalisations.

The framework issues predictions that the programme has been unable to locate in any theory confined to a single domain. It predicts the scaling behaviour of *new* recursive systems from their composition operator alone. It predicts a scaling crossover at a derived depth R^* that shifts systematically with base quality. It predicts that measured β determines α via $\alpha = 1/(1 - \beta)$, a relationship validated computationally to $R^2 = 1.00000000$. It derives the ARC Bound ($\beta = 0.5$, $\alpha = 2$) as the proposed stable-scaling limit for classical sequential computation, grounded in the pacing derivation of Law III at $\gamma = 1/2$ (correction 9), the $O(N^2)$ tally of self-attention pathways serving only as a heuristic for the frozen regime: a scaling limit, not a speed limit; the most a recursively self-correcting system can grow whilst remaining correctable; systems can exceed it in transient supercritical excursions but do not stay stably self-correcting there; and architecture-bound, so the $O(N^2)$ basis of the derivation is expiring rather than permanent, and a successor architecture beyond pairwise attention would relocate the limit rather than remove the constraint. It is a derived structural constraint, not an empirical coincidence. And it predicts that bounded systems exhibit composition operator transitions that explain why different studies of the same system can observe different scaling regimes (Theorem 5). These predictions are specific, falsifiable, and forward-looking.

Update: Revised Empirical Standing

The v5 cross-architecture experiment (Paper II, March 2026) has tested the AI predictions against six frontier models. The results require an honest reassessment of what is confirmed and what requires revision:

Confirmed: (1) The sequential advantage is universal: $\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model tested. This is the framework's core structural prediction for AI, and it holds without exception. (2) The power-law functional form fits well for at least one model (Gemini 3 Flash, $r^2 = 0.86$), consistent with multiplicative composition. (3) Error reduction with increased sequential compute investment is real.

Requires revision: (1) The AI scaling exponent is $\alpha_{\text{compute}} \approx 0.49$ (best fit), not ≈ 2 . The framework predicted super-linear scaling ($\alpha > 1$); the data shows sub-linear scaling. The key question has shifted from 'can α exceed the quadratic bound?' to 'can α reliably exceed 1?' (2) α is architecture-dependent (ranging from ~ 0 to ~ 3 across models), not universal. (3) Alignment scaling (α_{align}) is architecture-dependent, ranging from -0.25 to $+0.44$, exhibiting a three-tier hierarchy (3 positive, 2 flat, 1 negative) rather than being universally near zero.

Open: The quadratic bound conjecture ($\alpha \leq 2$) is neither confirmed nor refuted. The β -to- α derivation, the R^* crossover, and the composition operator classification remain mathematically valid but empirically untested in physical systems. The cross-domain predictions (biology, physics) are unaffected by the AI results.

The mathematical framework survives intact. What has changed is the empirical picture for AI systems: the recursive advantage is real but weaker than predicted. Whether this reflects fundamental limitations of current architectures or limitations of the experimental design (problem difficulty, compute range, measurement methodology) remains to be determined.

If the predictions fail, we will have learned where the structural analogy breaks down. If they hold, we will have identified a principle that connects intelligence, computation, and the physics of order across substrates.

Intelligence may not be a property of particular materials. It may be what happens on the far side of the recursive threshold.

The predictions are specified. The falsification criteria are public. The data will decide.

APPENDIX A: MATHEMATICAL PROOFS

A.1 Proof of Theorem 1 (Classification)

Case 1: Multiplicative composition

Hypothesis: $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ continuous with $g(1) = 1$ and $g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$.

Define $h(x) = \ln g(e^x)$. Then:

$$h(x + y) = \ln g(e^{x+y}) = \ln g(e^x \cdot e^y) = \ln[g(e^x) \cdot g(e^y)] = h(x) + h(y)$$

This is Cauchy's additive functional equation. Under the continuity of g (hence h), the unique solution is $h(x) = \alpha x$ for some constant α . Therefore $\ln g(e^x) = \alpha x$, giving $g(R) = R^\alpha$. ■

Case 2: Additive composition

Hypothesis: $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ continuous with $f(0) = 1$ and $f(R_1 + R_2) = f(R_1) \cdot f(R_2)$.

Taking logarithms: $\ln f(R_1 + R_2) = \ln f(R_1) + \ln f(R_2)$. This is again Cauchy's additive equation, with unique continuous solution $\ln f(R) = \alpha R$, giving $f(R) = e^{\alpha R}$. ■

Case 3: Bounded composition

$Q_{r+1} = \min(Q_r + \delta Q_r, Q_{\max})$. For any monotonically increasing sequence $\{Q_r\}$ bounded above by $Q_{\max}$, $\lim_{r \rightarrow \infty} Q_r \leq Q_{\max}$. Therefore $f(R) = Q(R)/I$ is bounded, and U saturates. ■

A.2 Proof of Theorem 2 (Scaling Exponent)

From Axiom 2: $dg/dr = ag^\beta$, $\beta \in [0, 1)$, $g(0) = 1$.

Separating variables:

$$\begin{aligned} \int_1^g s^{-\beta} ds &= \int_0^R a dr \\ \frac{g^{1-\beta} - 1}{1-\beta} &= aR \\ g(R) &= [1 + (1-\beta)aR]^{1/(1-\beta)} \end{aligned}$$

For $R \gg R^*$ (where $(1-\beta)aR \gg 1$):

$$g(R) \approx [(1-\beta)aR]^{1/(1-\beta)} \propto R^{1/(1-\beta)}$$

Therefore $\alpha = 1/(1-\beta)$. ■

A.3 Proof of Theorem 3 (Crossover Depth)

The full solution $g(R) = [1 + (1 - \beta)aR]^{1/(1-\beta)}$ can be analysed by defining $\rho = (1 - \beta)aR$:

$$g(R) = (1 + \rho)^{1/(1-\beta)}$$

For $\rho \ll 1$ (small R): $g \approx 1 + \rho/(1 - \beta) = 1 + aR$, approximately linear in R .

For $\rho \gg 1$ (large R): $g \approx \rho^{1/(1-\beta)} \propto R^\alpha$, power law.

The crossover occurs at $\rho \approx 1$, giving:

$$R^* = \frac{1}{(1 - \beta)a} = \frac{\alpha}{a}$$

R^* depends only on α and a , both independently measurable. It is independent of base potential I , consistent with the separability of Axiom 1. ■

A.4 Uniqueness of Axiom 2

Why $dg/dr = ag^\beta$ and not some other functional form? This is the Bernoulli-type ODE for cumulative advantage (also known as preferential attachment), applied to the amplification factor g rather than absolute capability. The exponent β parameterises the degree of self-reference: $\beta = 0$ gives constant growth (no feedback); $\beta \rightarrow 1$ gives explosive growth (maximal feedback). The choice $\beta \in [0, 1)$ is the minimal model for a system where accumulated amplification accelerates future gains without divergence at finite R . More complex models (e.g., $dg/dr = ag^\beta - bg^\gamma$ with damping) would introduce additional parameters; we adopt the minimal form and test it against observation.

APPENDIX B: MEASUREMENT PROTOCOL

To test the framework, we propose a standardised six-step protocol for any recursive system:

Step 1. Define one recursive cycle (one self-referential step where output becomes input).

Step 2. Measure base capability I at $R = 1$ (no recursion). For cross-domain comparability, we propose operationalising I as the maximum rate of entropy reduction per recursive cycle, measured in bits per cycle: $I = \max_R [-\Delta H / \Delta R]$ evaluated at the first cycle ($R = 0$ to $R = 1$). This connects to established physics through Landauer's principle (1961): each $kT \ln 2$ of free energy processed corresponds to one bit of information. In AI, I is the accuracy improvement from zero reasoning to one step. In quantum error correction, I is the error reduction from one syndrome cycle. In biology, I is the metabolic efficiency gain from one branching level (in thermodynamic bits via the Landauer bridge). This definition gives $U = I \times R^\alpha$ consistent units: bits = (bits/cycle) $\times$ (cycles) $^\alpha$.

Step 3. Measure capability U at minimum 5 recursive depths spanning one order of magnitude. Look for the R^* crossover.

Step 4. Fit all three models: power law ($\log(U/I) = \alpha \log R$), exponential ($\log(U/I) = \lambda R$), logarithmic ($U/I = k \log R$). Select best fit via AIC/BIC. The power law is a prediction to test, not an assumption.

Step 5. Measure β independently: plot $\log(\Delta U)$ against $\log(U_{\text{accumulated}})$ at each depth; the slope estimates β . Verify whether $\alpha \approx 1/(1 - \beta) \pm 0.3$.

Step 6. Report α with 95% confidence intervals. Submit to public repository.

APPENDIX C: COMPUTATIONAL VALIDATION SUITE

This appendix summarises the results of a comprehensive computational validation programme testing Theorems 1–5. All validation code is available as supplementary material. Tests were conducted using Python 3.12 with NumPy, SciPy, and Matplotlib.

C.1 Theorem 1 Validation (Cauchy Classification)

Fifteen systems were tested across three composition regimes (5 multiplicative, 5 additive, 5 bounded). Each system was evaluated on whether its measured composition rule matched the predicted scaling function.

Result: 15/15 correct classifications (100%). No false positives (non-recursive systems were not classified as recursive), no false negatives. The Cauchy functional equation classification is exhaustive and exact under the stated conditions.

C.2 Theorem 2 Validation ($\beta \rightarrow \alpha$ Identity)

The relationship $\alpha = 1/(1 - \beta)$ was tested against 30 exact Bernoulli ODE solutions with β spanning 0.05 to 0.92. For each test case, β was measured blindly from the marginal gain structure (dg/dR vs g in log-log space), α was predicted, and the predicted value was compared against the true scaling exponent.

Result: $R^2 = 1.00000000$ (eight decimal places). Regression slope: 1.000102. Mean absolute error: 0.002%. This confirms $\alpha = 1/(1 - \beta)$ is an exact analytical identity, not an empirical approximation.

C.3 Theorem 2a Validation (Non-Additivity)

The non-additivity proof was verified numerically. For multiple (I_1, I_2, R_1, R_2) combinations across all $\beta \in (0, 1)$: if U were additive ($U = g(I) + h(R)$), then $U(I_1, R_1) - U(I_1, R_2)$ would equal $U(I_2, R_1) - U(I_2, R_2)$. In all cases, the differences are unequal, confirming the multiplicative interaction is necessary.

Result: Non-additivity confirmed for all tested parameter combinations. Synergy quotient $S > 1$ for all $\beta \in (0, 1)$ and all $R > 0$.

C.4 Theorem 3 Validation (R^* Crossover)

The crossover depth $R^* = 1/[(1 - \beta)a] = \alpha/a$ was tested by comparing the predicted crossover with the numerically measured transition point (where $\rho = 1$).

Result: Functional form confirmed. Absolute R^* values show 2–7% error, attributable to the approximation of ‘crossover at $\rho \approx 1$.’ The R^* independence from I is confirmed: varying base potential does not shift the crossover depth, consistent with the separability of Axiom 1.

C.5 Theorem 4 Validation (Five Properties)

Ten recursive systems and five non-recursive controls were scored on the five derived properties. Recursive systems scored a mean of 4.2/5; non-recursive controls scored 2.8/5. The difference ($\Delta = 1.4$) is consistent with the framework's prediction but the sample size yields $p = 0.095$, which does not reach conventional significance. *(In plain English: the result goes in the predicted direction, recursive systems do score higher, but with only 15 total samples, there is roughly a 1-in-10 chance this pattern could appear by chance. Scientists typically require 1-in-20 or better. More data is needed.)*

Result: Direction correct; statistical power insufficient for definitive confirmation. Larger samples are required.

C.6 Theorem 5 Validation ($\oplus$ Phase Transitions)

Four systems were tested for depth-dependent composition transitions: (1) exact Bernoulli ODE (control), (2) logistic growth, (3) gradient descent with momentum, (4) Kuramoto oscillators.

Results: The Bernoulli control maintained constant β ($\sigma < 10^{-6}$). All three bounded systems showed systematic β transitions: logistic growth from $\beta \approx 0.5$ (growth phase) to $\beta \rightarrow -\infty$ (saturation), gradient descent from $\beta \approx 0.52$ (exploration) to $\beta \approx 0.98$ (convergence), Kuramoto oscillators from $\beta \approx 1.7$ (growth) to $\beta \approx 20$ (saturation). Transitions are generic in bounded systems and absent in unbounded systems.

C.7 Negative Controls

Four non-recursive systems were tested: (1) Gaussian noise, (2) linear decay, (3) sinusoidal oscillation, (4) random walk. None produced false positive signals ($\mathcal{S} > 1$ or $\alpha > 1$).

Result: 0/4 false positives. The framework correctly identifies the absence of recursive amplification in systems that do not satisfy the axioms.

C.8 Honest Failures and Boundary Cases

Two results fell short of expectations:

R^* absolute value: 62% mean error in predicted vs actual crossover depth (functional form correct, proportionality constant off). This likely reflects sensitivity to the $\rho \approx 1$ approximation.

Newton's method: Local convergence near roots yields $\beta > 1.0$ (quadratic convergence), which falls outside the framework's domain $\beta \in [0, 1)$. The framework correctly identifies this as a boundary case where the Bernoulli ODE model does not apply.

These failures are reported for completeness and scientific integrity. They bound the framework's applicability and inform future refinement.

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Companion Documents

Eastwood, M.D. (2026). Paper III: The Alignment Scaling Problem. First published 9 February 2026. OSF DOI: 10.17605/OSF.IO/HQCGF.

Eastwood, M.D. (2026). Eden Protocol: Engineering Specification. First published 22 February 2026. OSF DOI: 10.17605/OSF.IO/AWJR4.

Eastwood, M.D. (2026). Eden Protocol: Philosophical Vision. First published 22 February 2026. OSF DOI: 10.17605/OSF.IO/9M3DG.

Eastwood, M.D. (2026). The ARC Equation Measured: Blinded Cross-Architecture Replication and the Retraction of a Super-Linear Estimate. Paper II. Current draft, version 2.11, revised 18 September 2026 (v5 cross-architecture results). First published 22 January 2026; the multi-model study extended March 2026. Six models, 30 problems, bootstrap CIs. OSF DOI: 10.17605/OSF.IO/8FJMA.

Eastwood, M.D. (2026). On the Origin of Scaling Laws. March 2026. Cross-domain accessible presentation of the ARC Principle, including the Completeness Theorem, Phase Diagram of Complexity, and Dimensional Ladder.

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v5.16 (21 September 2026): six dated corrections have been added to the author's note, numbered 14 to 19, and a scoping update has been added beneath the falsification status summary. Substantive: correction 14 separates Theorem 1's premises, with all three results standing as printed; correction 15 carries the added-recursion clock declared at Axiom 2 into the printed solution and into Appendix A; correction 16 names the added premises behind two of Theorem 4's five properties; correction 17 states that amplification is convex at every depth once the coupling is positive and that the crossover marks where compounding becomes numerically dominant; correction 18 states that Appendix A.1 case 3 establishes a bound rather than attainment, and gives the attainment condition; correction 19 records that the crossover is predicted to be independent of base quality, the conclusion's contrary clause being the residue of the formulation superseded at correction 1. Also substantive: the scoping update records that Paper II now frames the cross-architecture replication on five analysable models of the six tested. Editorial: Paper II is cited at its current version and revision date in the references. No theorem, result, figure or falsification criterion is withdrawn, no withdrawn material is restored or weakened into a replacement, and the first published date is unchanged. v5.15: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

THE ARC PRINCIPLE: FOUNDATIONAL PAPER - Current draft

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The predictions are specified. The falsification criteria are public. The data will decide.

Companion documents: **On the Origin of Scaling Laws** (accessible cross-domain paper) | **Paper III** (extended evidence, experimental protocols, blind prediction test results) | **Eden Protocol** (alignment architecture specification) | **Philosophical Vision** | **Paper II** | **Paper V: The Stewardship Gene** | **Executive Summary**.

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